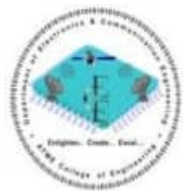




ATME
College of Engineering



Subject Name: Microwave Engineering & Antenna Theory

Subject Code: BEC701

Course Coordinator:

Guruprasad K N

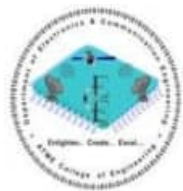
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ATME

College of Engineering



Module 3

Strip Lines & Antenna Basics

Strip Lines:

- Introduction
- Micro Strip lines
- Parallel Strip lines

Suggested Reading

Text 2: “Microwave Devices and Circuits” -
Liao, Pearson Education,

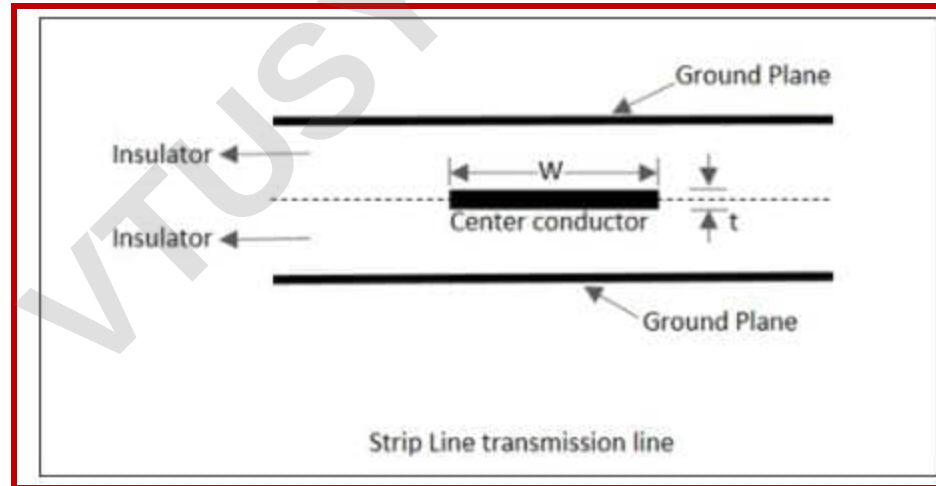
Text 2:

Sections: I I

Strip Lines:

These are the *planar transmission lines*, used at frequencies from 100MHz to 100GHz.

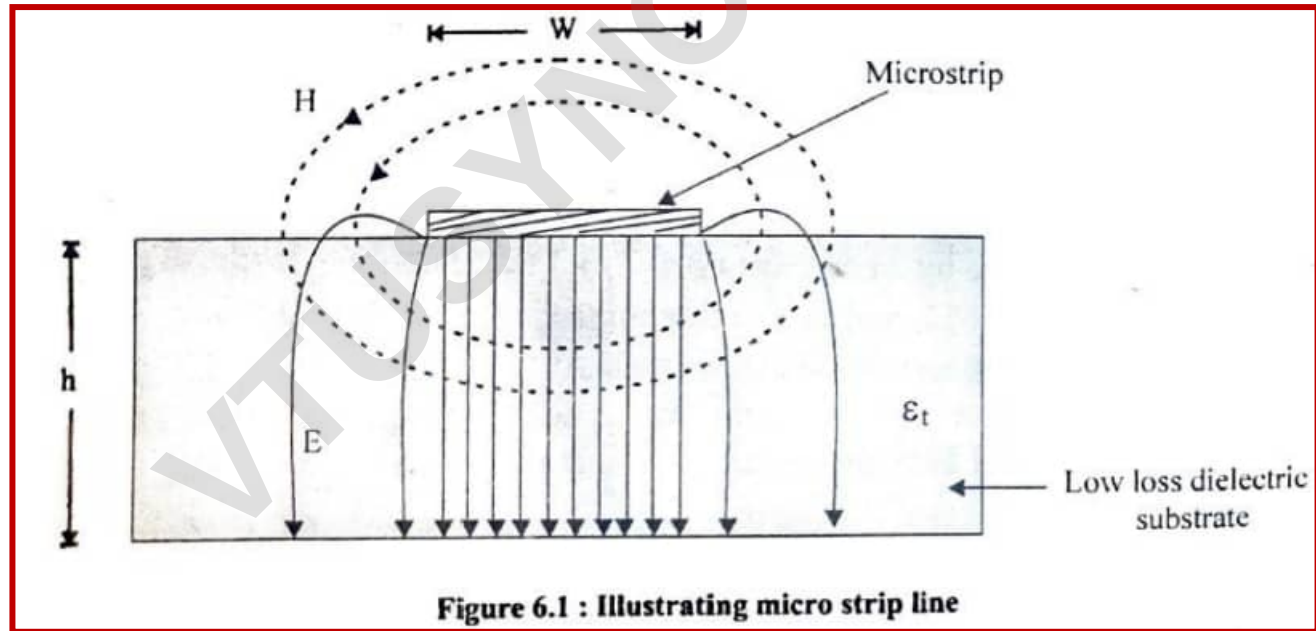
A **Strip line** consists of a central thin conducting strip of width w which is greater than its thickness t . It is placed inside the low loss dielectric (ϵ_r) substrate between two wide ground plates. The width of the ground plates is five times greater than the spacing between the plates.



All microwave equipments used co-axial lines, waveguides (both rectangular and cylindrical) or parallel strip line circuits for transmitting microwaves from one device to another till the year about 1965. Later on, development of planar transmission lines have resulted in miniaturization of microwave circuits. These planar transmission lines are flat having two or multiconductor transmission lines having low profile and light weight. The characteristic impedance of the line depends on the line dimensions and is hence suitable for microwave integrated circuits (MICs). By using **thin film technology** and **photolithographic techniques**, it is possible to fabricate the complete transmission line circuits. There are four basic types of planar transmission lines which are widely used in MICs. They are (a) microstrip lines, (b) parallel strip lines, (c) coplanar strip lines and (d) shielded strip lines.

6.2 MICROSTRIP LINES

Figure 6.1 shows a microstrip line consisting of a single ground plane and a thin strip conductor on a low loss dielectric substrate above the ground plane. The microwave solid state active devices can be easily mounted on top and easily connected to these microstrip lines.



Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019
Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing
ONE full question from each module.

Module-1

1. a. Derive the general transmission line equation to find voltage and current on the line in terms of position 'z' and time 't'. (07 Marks)
- b. Describe the different mode curve in the case of reflex klystron. (05 Marks)
- c. A transmission line has a characteristic impedance of $50 + j0.01\Omega$ and terminated in a load impedance of $73 - j42.5\Omega$ calculate : i) reflection coefficient ii) SWR. (04 Marks)

OR

2. a. Define reflection coefficient. Derive the equation for reflection coefficient at the load end at a distance 'd' from the load. (06 Marks)
- b. Describe the mechanism of oscillation of reflex klystron. (06 Marks)
- c. A transmission line has the following parameters : $R = 2\Omega/m$, $G = 0.5\text{mmho/m}$, $f = 1\text{GHz}$, $L = 8\text{nH/m}$, $C = 0.23\text{pF/m}$. Calculate : i) characteristic impedance ii) propagation constant. (04 Marks)

Module-2

3. a. State and explain the properties of S - matrix. (07 Marks)
- b. With a neat diagram, explain the working of precession type variable attenuator. (06 Marks)
- c. A 20mW signal is fed into one of the collinear port 1 of a lossless H-plane T junction. Calculate the power delivered through each port when other ports are terminated in matched load. (03 Marks)

OR

4. a. What is magic Tee? Derive its scattering matrix. (06 Marks)
- b. Discuss different types of coaxial connectors. (04 Marks)
- c. 2 transmission lines of characteristic impedance Z_1 and Z_2 are joined at plane PP'. Express S-parameters in terms of impedance when each line is matched terminated. (06 Marks)

Module-3

5. a. Explain the construction and field pattern for microstrip line. (06 Marks)
- b. Explain the following terms as related to antenna system :
i) directivity ii) beam efficiency iii) effective aperture. (06 Marks)

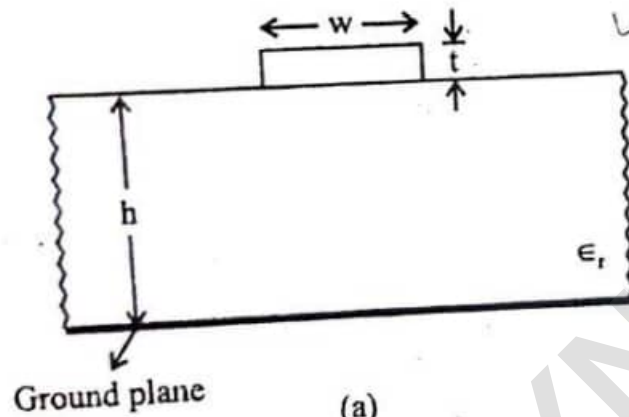
Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal in evaluator and/or equations written eg. 42-8-50, will be treated as malpractice.

The electric field lines remain partially in air and partially in the lower dielectric substrate as shown in figure 6.1. This makes the mode of propagation not pure TEM but quasi-TEM. Since it is an open structure, microstrip lines radiate electro-magnetic energy. The radiation loss is proportional to the square of the frequency. By using thin and high dielectric materials, the radiation losses can be reduced.

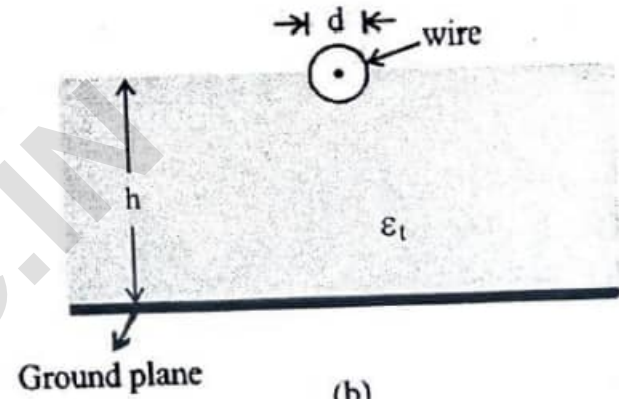
Characteristic Impedance Z_0 of Microstrip line

Characteristic Impedance Z_0 of Microstrip Lines

The characteristic impedance of a microstrip line depends on the width " w " of the strip line, the thickness " t " of the strip line, the distance " h " between the stripline and ground plane and the effective homogeneous dielectric constant ϵ_{re} of the dielectric substrate. Different methods have been developed for determining the characteristic impedance of a microstrip line in terms of the above mentioned parameters w , t , h and ϵ_{re} . These methods are extremely complicated. A fairly easy way of determining Z_0 is to compare it with the characteristic impedance of wire-over-ground line and suitably modifying Z_0 . For this let us consider a microstrip line of width " w ", thickness " t " mounted on a dielectric slab of thickness " h " having dielectric constant " ϵ_r " as shown in fig. 6.2(a) and a wire-over-ground line of diameter ' d ' with its axis at a height ' h ' over ground line. The space between wire and ground is filled with a dielectric material having same dielectric constant as the substrate ϵ_r as shown in fig. 6.2(b).



(a)



(b)

Figure 6.2 : (a) Microstrip line

(b) Wire-over-ground line

The characteristic impedance of a wire-over-ground transmission line is given by

$$Z_0 = \frac{60}{\sqrt{\epsilon_r}} \ln \frac{4h}{d} \quad \dots (6.1)$$

The characteristic impedance Z_0 of the microstrip line can be calculated using a similar equation as (6.1) but with modified values for dielectric constant ϵ_r and the diameter d .

Effective Dielectric Constant ϵ_{re}

The effective dielectric constant ϵ_{re} for a microstrip line is related to the relative dielectric constant through an empirical relation developed by the researcher DiGiacomo given by

$$\epsilon_{re} = 0.475 \epsilon_r + 0.67 \quad \dots (6.2)$$

where

ϵ_r = relative dielectric constant

ϵ_{re} = effective relative dielectric constant for a microstrip line

Expression for Diameter d of the Wire-over-ground

The rectangular conductor of the microstrip line has to be transformed into an equivalent circular conductor so that the diameter d of the circular conductor can be expressed in terms of 'w' and 't'. Another researcher Springfield obtained one more empirical relationship connecting 'd', 'w' and 't' given by

$$d = 0.67w \left(0.8 + \frac{t}{w} \right) \quad \dots (6.3)$$

The above relation holds good for the ratio $\left(\frac{h}{w} \right)$ varying from 0.1 to 0.8.

Using equations (6.3) and (6.2) in (6.1), we get the characteristic impedance Z_0 of microstrip line as

$$Z_0 = \frac{60}{\sqrt{0.475 \epsilon_r + 0.67}} \ln \frac{4h}{0.67w \left(0.8 + \frac{t}{w}\right)}$$

$$= \frac{60/\sqrt{0.475}}{\sqrt{\epsilon_r + \frac{0.67}{0.475}}} \ln \left[\frac{(4/0.67)h}{(0.8w + t)} \right]$$

$$\therefore Z_0 = \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left[\frac{5.98 h}{0.8w + t} \right] \text{ for } h < 0.8 w \quad \dots (6.4)$$

The velocity of wave propagation is given by

$$v = \frac{c}{\sqrt{\epsilon_{re}}} = \frac{3 \times 10^{10}}{\sqrt{\epsilon_{re}}} \text{ cm/sec} \quad \dots (6.5)$$

For a wide strip line (i.e., $w \gg h$), the characteristic impedance was calculated to be

$$Z_0 = \frac{h}{w} \sqrt{\frac{\mu}{\epsilon}} = \frac{377}{\sqrt{\epsilon_r}} \frac{h}{w} \quad \dots (6.6)$$

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1.
 - a. List four applications of Reflex Klystron. (04 Marks)
 - b. Derive transmission line equations in voltage and current forms. (06 Marks)
 - c. A transmission line is terminated in a resistive load of 1000Ω and has $L = 9\mu\text{H/m}$ and $C = 100\text{pF/m}$. Calculate reflection coefficient and standing wave ratio. (06 Marks)

OR

2.
 - a. Define reflection coefficient. Derive an expression for reflection coefficient at load in terms of characteristic impedance and load impedance. (08 Marks)
 - b. Explain microwave system with the aid of a diagram. (08 Marks)

Module-2

3.
 - a. For a two port network with mismatched load derive an expression for input reflection coefficient. (06 Marks)
 - b. Draw the diagram of Magic-Tee. Derive S-matrix of Magic Tee. (10 Marks)

OR

4.
 - a. What is a reciprocal device? Write five point comparison among [S], [Z] and [Y] matrices. (06 Marks)
 - b. Given $[Z] = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ Find S-matrix. (05 Marks)
 - c. Explain coaxial line fixed attenuator with a diagram. (05 Marks)

Module-3

5.
 - a. Derive characteristic impedance of micro-strip lines. (08 Marks)
 - b. Define the following terms with respect to antennas :

Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Scaling of identification appeal to evaluator and/or equations written eg. 42-18-30, will be treated as malpractice.

Example 6.1 : Calculate the characteristic impedance Z_0 of a microstrip line given that the thickness of the dielectric substrate $h = 175$ microns, thickness of the strip $= 70$ microns, the width $w = 250$ microns and the dielectric constant of the substrate material is 4.5.

Since h is less than $0.8 w$ ($= 0.8 \times 250 = 200 \mu\text{m}$).

$$Z_0 = \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left[\frac{5.98 h}{0.8 w + t} \right]$$

Example 6.1 : Calculate the characteristic impedance Z_0 of a microstrip line given that the thickness of the dielectric substrate $h = 175$ microns, thickness of the strip $= 70$ microns, the width $w = 250$ microns and the dielectric constant of the substrate material is 4.5.

Solution : Since h is less than $0.8 w$ ($= 0.8 \times 250 = 200 \mu\text{m}$).

Equation 6.4 can be used,

$$\begin{aligned} \therefore Z_0 &= \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left[\frac{5.98 h}{0.8 w + t} \right] \\ &= \frac{87}{\sqrt{4.5 + 1.41}} \ln \left[\frac{(5.98) (175 \times 10^{-6})}{0.8 \times 250 \times 10^{-6} + 70 \times 10^{-6}} \right] \\ \therefore Z_0 &= 48.48 \Omega \end{aligned}$$

Example 6.2 : Calculate the characteristic impedance of a wide microstrip line having negligible thickness and having a width of 0.8 mm. The thickness of the substrate is 0.2 mm and has a dielectric constant of 3.55.

Solution : Given $w = 0.8$ mm, $h = 0.2$ mm, $t \simeq 0$, $\epsilon_r = 3.55$.

Since $w \gg h$, the characteristic impedance given by

$$Z_0 = \frac{377}{\sqrt{\epsilon_r}} \left(\frac{h}{w} \right)$$

Example 6.2 : Calculate the characteristic impedance of a wide microstrip line having negligible thickness and having a width of 0.8 mm. The thickness of the substrate is 0.2 mm and has a dielectric constant of 3.55.

Solution : Given $w = 0.8$ mm, $h = 0.2$ mm, $t \simeq 0$, $\epsilon_r = 3.55$.

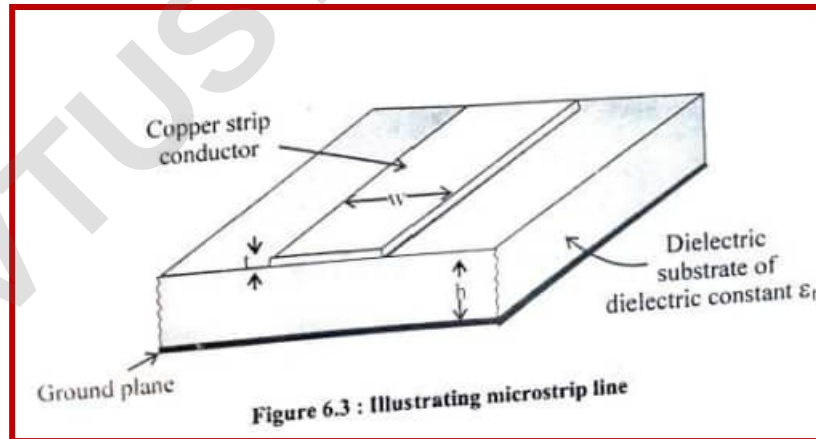
Since $w \gg h$, equation (6.6) is used for calculating the characteristic impedance given by

$$Z_0 = \frac{377}{\sqrt{\epsilon_r}} \left(\frac{h}{w} \right) = \frac{377}{\sqrt{3.55}} \left(\frac{0.2}{0.8} \right)$$

$$\therefore Z_0 = 50 \Omega$$

LOSSES IN MICROSTRIP LINES

Let us consider a microstrip line consisting of a conductive copper strip attached to a dielectric sheet with ground plane backing as shown in fig. 6.3. This type of transmission line is widely used in microwave integrated circuits and computer technology. In addition to the two characteristics of a microstrip line namely characteristic impedance and velocity of wave propagation, there is another characteristic of the line namely **attenuation**. The attenuation constant of microstrip line depends on frequency of operation, electrical properties of substrate and the conductors (both microstrip and ground plane) and the geometry of mounting of strip on the dielectric.



When the dielectric substrate of dielectric constant ϵ_r is purely non-magnetic, then three types of losses occur in microstrip lines. They are

- (a) Dielectric losses in the substrate
- (b) Ohmic losses in the strip conductor and the ground plane
- (c) Radiation losses

CBCS SCHEME

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15EC71

Seventh Semester B.E. Degree Examination, June/July 2019 Microwave and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-3

- 5 a. A lossless parallel strip line has a conducting strip width W . The substrate dielectric separating the two conducting strips has a relative dielectric constant ϵ_{rd} of 6 and a thickness d of 4mm. Calculate: i) The required width W of the conducting strip in order to have a characteristic impedance of 50Ω ; ii) The strip-line capacitance. (04 Marks)
- b. Discuss different types of losses in microstrip lines. (06 Marks)
- c. Calculate the exact directivity for 3 dimensional source having the pattern $U = U_m \sin^2 \theta \sin^3 \phi$ where $0 \leq \theta \leq \pi$, $0 \leq \phi \leq \pi$. (06 Marks)

CBCS SCHEME

USN

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15EC71

Seventh Semester B.E. Degree Examination, Dec.2019/Jan.2020

Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-3

- 5 a. Derive characteristic impedance of micro-strip lines. (08 Marks)
b. Define the following terms with respect to antennas :
i) Beam area
ii) Radiation intensity
iii) Beam efficiency
iv) Directivity. (08 Marks)

OR

- 6 a. Describe ohmic skin losses and radiation losses in micro-strip lines. (10 Marks)
b. A parabolic reflector antenna is circular in cross section with a diameter of 1.22m. If the maximum effective aperture is 55% of the physical aperture, calculate gain of the antenna in dB at 20 GHz. (06 Marks)

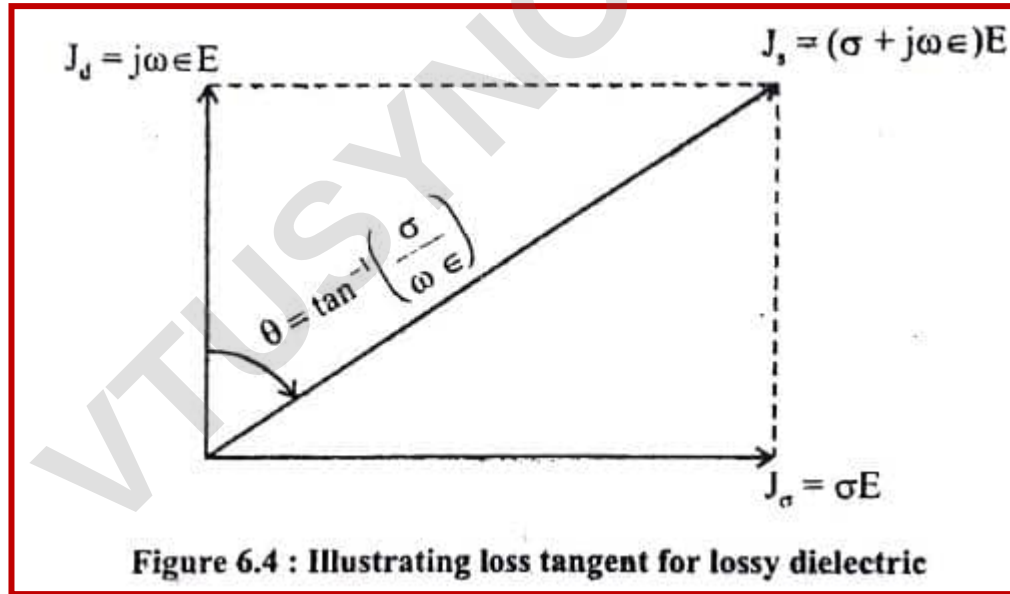
The amount of electric current travelling per unit cross-section area is called as **current density** and expressed in amperes per square meter.

Difference between the conduction current density and displacement current density?

Conduction current density is defined when the charge carriers are electrons. This is the current that is mostly talked about. Conduction current arises when there are free electrons in a substance that are not too rigidly bound to the atom. Such electrons are available in plenty in conductors like copper. The total amount of charge passing a point per second per unit area is then the conduction current density. On the other hand, the displacement current density arises from electrons that are bound to the atom and are not free to move. Such electrons are displaced slightly when subjected to an electric field. Displacement current density exists in the presence of alternating electric fields and is proportional to the frequency of the field. In the Maxwell's equation $\text{curl } \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t = \sigma \mathbf{E} + i\omega \epsilon \mathbf{E}$ where $\mathbf{J}$ is conduction current density and $\partial \mathbf{D} / \partial t$ is displacement current density ($\mathbf{D}$ is electric displacement having dimensions of charge per unit area), $\mathbf{E}$ is electric field, σ is conductivity, ϵ is electric permittivity and ω is angular frequency. The displacement current density is negligibly small at low frequencies.

(a) Dielectric Losses

All dielectric materials possess some conductivity σ but it will be small such that $\sigma < \omega\epsilon$. When this conductivity is not negligible, then the displacement current density leads the conduction current density by 90° , thus introducing loss tangent for a lossy dielectric*, as shown in fig. 6.4.



The dielectric attenuation constant is given by

$$\alpha_d = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} \text{ nepers/cm} \quad \dots (6.7)$$

where σ = conductivity of the dielectric substrate in Ω/cm .

From figure 6.4 the dielectric loss tangent is given by.

$$\tan \theta = \frac{\sigma}{\omega \epsilon} \quad \dots (6.8)$$

Eliminating σ in equations (6.7) and (6.8), we get

$$\alpha_d = \frac{\omega \epsilon \tan \theta}{2} \sqrt{\frac{\mu}{\epsilon}}$$

$$\therefore \alpha_d = \frac{\omega}{2} \sqrt{\mu \epsilon} \cdot \tan \theta \text{ nepers/cm} \quad \dots (6.9)$$

The space above the microstrip is air where there are no dielectric losses, but below the microstrip, there is non-magnetic dielectric substrate. Due to this mix up, equation (6.9) has to be modified. This modification was done by researchers Welch, Pratt and later on by Pucel to obtain an empirical relationship for α_d as

$$\alpha_d = 1.634 \times 10^3 \frac{q\sigma}{\sqrt{\epsilon_{re}}} \text{ dB/cm} \quad \text{..... (6.10)}$$

where 1 neper = 8.686 dB

ϵ_{re} = effective dielectric constant of substrate as given by equation (6.2)

$$q = \text{dielectric filling factor} = \frac{\epsilon_{re} - 1}{\epsilon_r - 1} \quad \text{..... (6.11)}$$

Substituting for σ from equation (6.8) into equation (6.10), we get

$$\alpha_d = 1.634 \times 10^3 \frac{q\omega \epsilon \tan \theta}{\sqrt{\epsilon_{re}}} \quad \text{..... (6.12)}$$

we have $\omega = 2\pi f = 2\pi \frac{c}{\lambda_0} = \frac{2\pi c}{\lambda_g \sqrt{\epsilon_{re}}}$ (6.13)

where $\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{re}}} = \text{guide wavelength in the dielectric in cm}$

$\lambda_0 = \text{free space wavelength in cm}$

Using equation (6.13) in (6.12) and replacing ϵ by $\epsilon_0 \epsilon_r$, we get

$$\alpha_d = 1.634 \times 10^3 \times q \times \left(\frac{2\pi c}{\lambda_g \sqrt{\epsilon_{re}}} \right) \left(\frac{\epsilon_0 \epsilon_r}{\sqrt{\epsilon_{re}}} \right) \tan \theta$$

Substituting for $c = 3 \times 10^8$ m/sec, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m, we get

$$\alpha_d = 1.634 \times 10^3 \times 2\pi \times 3 \times 10^8 \times 8.854 \times 10^{-12} \left(\frac{q \epsilon_r}{\epsilon_{re}} \right) \frac{\tan \theta}{\lambda_g} \text{ dB/cm}$$

$$\therefore \alpha_d = 27.3 \left(\frac{q \epsilon_r}{\epsilon_{re}} \right) \frac{\tan \theta}{\lambda_g} \text{ dB/cm} \quad \text{..... (6.14)}$$

(b) Ohmic Losses

In a microstripline, the major contribution to losses at microwave frequencies is from the finite conductivity of the microstrip conductor placed on a low-loss dielectric substrate. Due to the current flowing through the strip, there will be ohmic losses and hence attenuation of the microwave signal takes place. The current distribution in the transverse plane is fairly uniform with minimum value at the central axis and shooting up to maximum values at the edge of the strip. Assuming uniform current distribution in the region $-\frac{w}{2} \leq x \leq \frac{w}{2}$, the attenuation constant due to ohmic loss of a wide microstrip line $\left(\frac{w}{h} > 1\right)$ is given by

$$\alpha_c \approx \frac{8.686 R_s}{Z_0 w} \text{ dB/cm for } \frac{w}{h} > 1 \quad \text{..... (6.15)}$$

where $R_s = \sqrt{\frac{\pi f \mu}{\sigma}} = \text{surface skin resistance in } \Omega/\text{cm}^2 \quad \text{..... (6.16)}$

Substituting for R_s , we get

$$\alpha_c \approx \frac{8.686}{Z_0 w} \sqrt{\frac{\pi f \mu}{\sigma}} \text{ dB/cm for } \frac{w}{h} > 1 \quad \text{..... (6.17)}$$

(c) Radiation Losses

At microwave frequencies, the microstrip line acts as an antenna radiating a small amount of power, resulting in radiation losses. This loss depends on the thickness of the substrate, the characteristic impedance Z_0 , the effective dielectric constant and the frequency of operation. Researcher Lewin showed that the ratio of radiated power P_{rad} to the total dissipated power P_t for an open microstrip is given by

$$\frac{P_{\text{rad}}}{P_t} = \frac{R_r}{Z_0} \quad \dots (6.18)$$

where

Z_0 = characteristic impedance

R_r = radiation resistance of the microstrip given by

$$R_r = 240 \pi^2 \left(\frac{h}{\lambda_0} \right)^2 F(\epsilon_{\text{re}})$$

where

λ_0 = Free-space wavelength

$F(\epsilon_{\text{re}})$ = radiation factor given by

$$F(\epsilon_{\text{re}}) = \frac{\epsilon_{\text{re}} + 1}{\epsilon_{\text{re}}} - \frac{\epsilon_{\text{re}} - 1}{2(\epsilon_{\text{re}})^{3/2}} \ln \frac{\sqrt{\epsilon_{\text{re}}} + 1}{\sqrt{\epsilon_{\text{re}}} - 1} \quad \dots (6.20)$$

Quality Factor Q of Microstrip Lines : The quality factor of microstrip lines depends on the dielectric attenuation constant α_d and conducting attenuation constant α_c . Usually the quality factor of a microstrip will be very high. But it is limited by the low dielectric constant of the substrate and the radiation losses of the substrates.

The conductor quality factor Q_c is related to the conductor attenuation constant by

$$Q_c = \frac{27.3}{\alpha_c} \quad \dots (6.21)$$

For a wide microstrip line, α_c is given by equation (6.15) as

$$\alpha_c \approx \frac{8.686 R_s}{Z_0 w}$$

Substituting, we get

$$Q_c = \frac{27.3 Z_0 w}{8.686 R_s} \quad \dots (6.22)$$

For a wide microstrip line, the characteristic impedance Z_0 is given by equation (6.6) as

$$Z_0 = \frac{377}{\sqrt{\epsilon_r}} \left(\frac{h}{w} \right) \quad \dots (6.23)$$

The wavelength in the microstrip line is given by

$$\begin{aligned} \lambda_g &= \frac{\lambda_0}{\sqrt{\epsilon_r}} \\ &= \frac{c}{f \sqrt{\epsilon_r}} \\ &= \frac{3 \times 10^{10}}{f_G \times 10^9 \sqrt{\epsilon_r}} \text{ cm} \\ \lambda_g &= \frac{30}{f_G \sqrt{\epsilon_r}} \\ \therefore \sqrt{\epsilon_r} &= \frac{30}{\lambda_g f_G} \quad \dots (6.24) \end{aligned}$$

where f_G = operating frequency in GHz

Using equation (6.24) in (6.23), we get

$$Z_0 = \frac{377 \lambda_g f_G}{30} \left(\frac{h}{w} \right) \quad \dots (6.25)$$

Using equation (6.24) in (6.23), we get

$$Z_0 = \frac{377 \lambda_g f_G}{30} \left(\frac{h}{w} \right) \quad \dots (6.25)$$

Using equation (6.25) in (6.22), we get

$$Q_c = \frac{27.3 w}{8.686 R_s} \left[\frac{377 \lambda_g f_G}{30} \left(\frac{h}{w} \right) \right]$$

$$Q_c = 39.5 \left(\frac{h}{R_s} \right) f_G \lambda_g$$

when α_c is expressed in terms of dB/λ_g , then the quality factor Q_c is expressed by

$$Q_c = 39.5 \left(\frac{h}{R_s} \right) f_G \quad \dots (6.26)$$

From equation (6.16), the surface skin resistance is given by

$$R_s = \sqrt{\frac{\pi f \mu}{\sigma}} = \sqrt{\frac{\pi f \mu_0 \mu_r}{\sigma}}$$

$$= \sqrt{\frac{\pi f_G \times 10^9 \times 4\pi \times 10^{-7} \times 1}{\sigma}}$$

$$\therefore R_s = 20\pi \sqrt{\frac{f_G}{\sigma}} \quad \text{..... (6.27)}$$

Using equation (6.27) in (6.26), we get

$$Q_c = 39.5 h f_G \left[\frac{1}{20\pi} \sqrt{\frac{\sigma}{f_G}} \right]$$

$$\therefore Q_c = 0.63 h \sqrt{\sigma f_G} \quad \text{..... (6.28)}$$

where h is measured in cm, f_G in GHz and σ is the conductivity of microstrip material in Ω/m .

In a similar way, the quality factor Q_d due to dielectric attenuation constant is given by

$$Q_d = \frac{27.3}{\alpha_d} \quad \dots (6.29)$$

where α_d is again expressed in dB/ λ_g .

From equation (6.14), we have

$$\alpha_d = 27.3 \left(\frac{q \epsilon_r}{\epsilon_{re}} \right) \frac{\tan \theta}{\lambda_g} \text{ dB/cm}$$

If α_d is expressed in terms of dB/ λ_g , then

$$\alpha_d = 27.3 \left(\frac{q \epsilon_r}{\epsilon_{re}} \right) \tan \theta \quad \dots (6.30)$$

Using equation (6.30) in (6.29), we get

$$Q_d = \frac{27.3}{(27.3) \left(\frac{q \epsilon_r}{\epsilon_{re}} \right) \tan \theta}$$

$$\therefore Q_d = \frac{\epsilon_{re}}{q \epsilon_r \tan \theta} \quad \dots (6.31)$$

From equation (6.11), $q = \frac{\epsilon_{re} - 1}{\epsilon_r - 1} \approx \frac{\epsilon_{re}}{\epsilon_r}$

Substituting, Q_d reduces to

$$Q_d \approx \frac{1}{\tan \theta} \quad \text{..... (6.32)}$$

Therefore, the Q_d for dielectric attenuation constant of a microstrip line is approximately the reciprocal of loss tangent $\tan \theta$ and is independent of frequency.

Example 6.3 : A microstrip line is composed of negligible thickness copper conductors mounted on a dielectric substrate of thickness 1.4 mm, loss tangent of 4×10^{-4} and dielectric constant of 9.6. The width of the microstrip line is 3 mm and operated at 10 GHz frequency. Determine

- (a) the characteristic impedance Z_0
 - (b) the effective dielectric constant
 - (c) attenuation due to dielectric losses
 - (d) attenuation due to conductor losses
 - (e) radiation losses if the total dissipated power is 4 mW
 - (f) quality factor Q_d due to dielectric attenuation
 - (g) quality factor Q_c due to conductor attenuation
- (Assume conductivity of copper as $\sigma = 5.8 \times 10^7 \text{ } \Omega^{-1}\text{m}^{-1}$).

Solution : Given $h = 1.4 \text{ mm} = 0.14 \text{ cm}$, $\tan \theta = 4 \times 10^{-4}$, $\epsilon_r = 9.6$, $w = 3 \text{ mm} = 0.3 \text{ cm}$,
 $f = 10 \times 10^9 \text{ Hz}$, $P_t = 4 \text{ mw}$.

(a) Since $w > h$,
the characteristic impedance is given by

$$Z_0 = \frac{377}{\sqrt{\epsilon_r}} \left(\frac{h}{w} \right)$$

(b) the effective dielectric constant ϵ_{re} is given by

$$\epsilon_{re} = 0.475 \epsilon_r + 0.67$$

Solution : Given $h = 1.4 \text{ mm} = 0.14 \text{ cm}$, $\tan \theta = 4 \times 10^{-4}$, $\epsilon_r = 9.6$, $w = 3 \text{ mm} = 0.3 \text{ cm}$, $f = 10 \times 10^9 \text{ Hz}$, $P_t = 4 \text{ mw}$.

- (a) Since $w > h$, the microstrip line can be considered to be wide and from equation (6.6), the characteristic impedance is given by

$$Z_0 = \frac{377}{\sqrt{\epsilon_r}} \left(\frac{h}{w} \right)$$

$$= \frac{377}{\sqrt{9.6}} \left(\frac{1.4}{3} \right)$$

$$\therefore Z_0 = 56.78 \Omega$$

- (b) From equation (6.2), the effective dielectric constant ϵ_{re} is given by

$$\epsilon_{re} = 0.475 \epsilon_r + 0.67$$

$$= (0.475) (9.6) + 0.67$$

$$\therefore \epsilon_{re} = 5.23$$

(c) the attenuation due to dielectric losses is given by

$$\alpha_d = 27.3 \left(\frac{q \epsilon_r}{\epsilon_{re}} \right) \frac{\tan \theta}{\lambda_g} \text{ dB/cm}$$

we have, $q = \frac{\epsilon_{re} - 1}{\epsilon_r - 1}$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{re}}} = \frac{c}{f \sqrt{\epsilon_{re}}}$$

(d) The attenuation due to conductor losses is given by

$$\alpha_c \approx \frac{8.686}{Z_0 w} \sqrt{\frac{\pi f \mu}{\sigma}} \text{ dB/cm}$$

$$\approx \frac{8.686}{Z_0 w} \sqrt{\frac{\pi f \mu_0 \mu_r}{\sigma}}$$

(c) From equations (6.14), the attenuation due to dielectric losses is given by

$$\alpha_d = 27.3 \left(\frac{q \epsilon_r}{\epsilon_{re}} \right) \frac{\tan \theta}{\lambda_g} \text{ dB/cm}$$

we have,

$$q = \frac{\epsilon_{re} - 1}{\epsilon_r - 1} = \frac{5.23 - 1}{9.6 - 1} = 0.492$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{re}}} = \frac{c}{f \sqrt{\epsilon_{re}}}$$

$$= \frac{3 \times 10^{10}}{10 \times 10^9 \sqrt{5.23}} = 1.312 \text{ cm}$$

Substituting, we get

$$\alpha_d = (27.3) \left(\frac{0.492 \times 9.6}{5.23} \right) \left(\frac{4 \times 10^{-4}}{1.312} \right)$$

$$\alpha_d = 7.5 \times 10^{-3} \text{ dB/m}$$

or

$$\alpha_d = 7.5 \times 10^{-3} \times 1.312 \text{ dB}/\lambda_g$$

or

$$\alpha_d = 9.86 \times 10^{-3} \text{ dB}/\lambda_g$$

(d) The attenuation due to conductor losses is given by equation (6.17) as

$$\begin{aligned}\alpha_c &\approx \frac{8.686}{Z_0 w} \sqrt{\frac{\pi f \mu}{\sigma}} \text{ dB/cm} \\ &\approx \frac{8.686}{Z_0 w} \sqrt{\frac{\pi f \mu_0 \mu_r}{\sigma}} \\ &\approx \frac{8.686}{(56.78)(0.3)} \sqrt{\frac{(\pi)(10 \times 10^9)(4\pi \times 10^{-7})(1)}{5.8 \times 10^7}}\end{aligned}$$

$$\therefore \alpha_c \approx 1.33 \times 10^{-3} \text{ dB/m}$$

$$\text{or } \alpha_c \approx 1.33 \times 10^{-3} \times 1.312 \text{ dB}/\lambda_g$$

$$\text{or } \alpha_c \approx 17.45 \times 10^{-3} \text{ dB}/\lambda_g$$

(e) the radiation factor is given by

$$F(\epsilon_{re}) = \frac{\epsilon_{re} + 1}{\epsilon_{re}} - \frac{\epsilon_{re} - 1}{2(\epsilon_{re})^{3/2}} \ln \frac{\sqrt{\epsilon_{re}} + 1}{\sqrt{\epsilon_{re}} - 1}$$

the radiation resistance is given by

$$R_r = 240 \pi^2 \left(\frac{h}{\lambda_0} \right)^2 F(\epsilon_{re})$$

Replacing λ_0 by $\frac{c}{f}$, we get

$$R_r = \frac{240 \pi^2 h^2 f^2 F(\epsilon_{re})}{c^2}$$

the radiated power is given by

$$P_{rad} = \frac{R_r}{Z_0} P_i$$

(e) From equation (6.20), the radiation factor is given by

$$F(\epsilon_{re}) = \frac{\epsilon_{re} + 1}{\epsilon_{re}} - \frac{\epsilon_{re} - 1}{2(\epsilon_{re})^{3/2}} \ln \frac{\sqrt{\epsilon_{re}} + 1}{\sqrt{\epsilon_{re}} - 1}$$

$$F(\epsilon_{re}) = \frac{5.23 + 1}{5.23} - \frac{5.23 - 1}{2(5.23)^{3/2}} \ln \frac{\sqrt{5.23} + 1}{\sqrt{5.23} - 1}$$

$$\therefore F(\epsilon_{re}) = 1.0254$$

From equation (6.19), the radiation resistance is given by

$$R_r = 240 \pi^2 \left(\frac{h}{\lambda_0} \right)^2 F(\epsilon_{re})$$

Replacing λ_0 by $\frac{c}{f}$, we get

$$R_r = \frac{240 \pi^2 h^2 f^2 F(\epsilon_{re})}{c^2}$$

$$= 240 \left(\frac{\pi \times 0.14 \times 10^{-3} \times 10 \times 10^9}{3 \times 10^{10}} \right)^2 (1.0254)$$

$$R_r = 5.29 \Omega$$

From equation (6.18), the radiated power is given by

$$P_{rad} = \frac{R_r}{Z_0} P_t = \frac{5.29}{56.78} \times 4 \text{ mw}$$

$$P_{rad} = 0.373 \text{ mw}$$

(f) the quality factor Q_d is given by

$$Q_d = \frac{\epsilon_{re}}{(q)(\epsilon_r) \tan \theta}$$

(g) the quality factor Q_c is given by

$$Q_c = 0.63 h \sqrt{\sigma f_G} \text{ where } h \text{ is in cm}$$

(f) From equation (6.31), the quality factor Q_d is given by

$$Q_d = \frac{\epsilon_{re}}{(q)(\epsilon_r) \tan \theta} = \frac{5.23}{(0.492)(9.6)(4 \times 10^{-4})}$$

$$Q_d = 2768.25$$

(g) From equation (6.28), the quality factor Q_c is given by

$$Q_c = 0.63 h \sqrt{\sigma f_G} \text{ where } h \text{ is in cm}$$

$$= (0.63)(0.14) \sqrt{5.8 \times 10^7 \times 10}$$

$$Q_c = 2124.14$$

6.3 PARALLEL STRIP LINES

Figure 6.4 shows a parallel strip lines consisting of two perfectly parallel strips made of copper. The space between the parallel strips is filled with a dielectric material having uniform thickness. The width of each strip is " w " and the thickness of the dielectric material is " d " and has a dielectric constant of ϵ_{rd} .

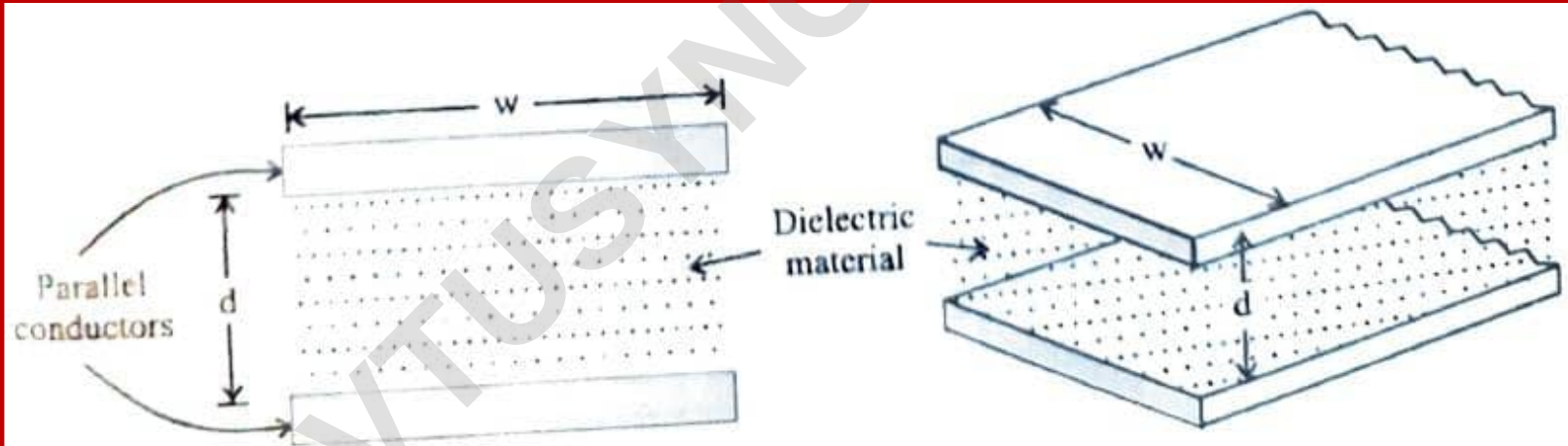


Figure 6.4 : Illustrating parallel strip lines

The mode of wave propagation is quasi-TEM mode in parallel strip lines which is similar to a two conductor system. If $w \gg d$, then the fringing capacitance becomes negligibly small. Then the inductance along the two conducting strips is given by

$$L = \frac{\mu_c d}{w} \text{ henry/m} \quad \dots (6.33)$$

where μ_c = permeability of the conductor

The capacitance between the two conducting parallel strips is given by

$$C = \frac{\epsilon_d w}{d} \text{ farad/m} \quad \dots (6.34)$$

The series resistance of both strips is given by

$$R = \frac{2R_s}{w} = \frac{2}{w} \sqrt{\frac{\pi f \mu_c}{\sigma_c}} \Omega/\text{m} \quad \dots (6.35)$$

where R_s = surface resistance

σ_c = conductivity of the strips in S/m

The shunt conductance of the parallel strip line is given by

$$G = \frac{\sigma_d w}{d} \text{ S/m} \quad \dots (6.36)$$

where σ_d = conductivity of the dielectric material between the two strips

Characteristic Impedance Z_0

The characteristic impedance of a lossless parallel stripline is given by

$$Z_0 = \sqrt{\frac{L}{C}} = \frac{d}{w} \sqrt{\frac{\mu_0 \mu_c}{\epsilon_0 \epsilon_{rd}}} = \frac{377}{\sqrt{\epsilon_{rd}}} \left(\frac{d}{w} \right) \Omega \quad \text{..... (6.37)}$$

The phase velocity of the TEM wave propagating through the parallel strip line is given by

$$v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\mu_0 \mu_c \epsilon_0 \epsilon_{rd}}} = \frac{c}{\sqrt{\epsilon_{rd}}} \text{ m/sec.} \quad \text{..... (6.38)}$$

Attenuation Losses

At microwave frequencies, the propagation constant of a parallel strip line can be expressed as

$$\gamma = \sqrt{ZY} = \sqrt{(R + j\omega L)(G + j\omega C)}$$

For $R \ll \omega L$ and $G \ll \omega L$ [Refer equations (1.50) & (1.51)] γ can be approximated by

$$\gamma = \frac{1}{2} \left[R \sqrt{\frac{C}{L}} + G \sqrt{\frac{L}{C}} \right] + j\omega \sqrt{LC} = \alpha + j\beta$$

Comparing real parts, the attenuation constant is given by

$$\alpha = \frac{1}{2} \left[R \sqrt{\frac{C}{L}} + G \sqrt{\frac{L}{C}} \right] \text{ nepers/m} \quad \dots (6.39)$$

And comparing imaginary parts,

$$\beta = \omega \sqrt{LC} \quad \dots (6.40)$$

$$\alpha = \frac{1}{2} \left[R \sqrt{\frac{C}{L}} + G \sqrt{\frac{L}{C}} \right] \text{ nepers/m} \quad \dots (6.39)$$

The attenuation constant for conductor losses is given by the 1st term of equation (6.39) as

$$\alpha_c = \frac{R}{2} \sqrt{\frac{C}{L}}$$

Substituting for R, C and L from respective equations (6.35), (6.34) and (6.33), we get

$$\alpha_c = \frac{1}{2} \cdot \frac{2}{w} \sqrt{\frac{\pi f \mu_c}{\sigma_c}} \sqrt{\frac{\epsilon_d w}{d} \cdot \frac{w}{\mu_c d}}$$

$$\therefore \alpha_c = \frac{1}{d} \sqrt{\frac{\pi f \epsilon_d}{\sigma_c}} \text{ nepers/m} \quad \dots (6.41)$$

And the attenuation constant for dielectric losses is given by the 2nd term of equation (6.39) as

$$\alpha_d = \frac{G}{2} \sqrt{\frac{L}{C}}$$

Again substituting for G, C and L from respective equations (6.36), (6.34) and (6.33), we get

$$\alpha_d = \frac{1}{2} \frac{\sigma_d w}{d} \sqrt{\frac{\mu_c d}{w} \cdot \frac{d}{\epsilon_d w}}$$

$$\therefore \alpha_d = \frac{1}{2} \sigma_d \sqrt{\frac{\mu_c}{\epsilon_d}}$$

$$= \frac{1}{2} \sigma_d \sqrt{\frac{\mu_0 \mu_{rc}}{\epsilon_0 \epsilon_{rd}}}$$

$$= \left(\frac{1}{2} \sigma_d \right) (377) \sqrt{\frac{1}{\epsilon_{rd}}} \text{ with } \mu_{rc} = 1 \text{ for conductor strips}$$

$$\therefore \alpha_d \simeq \frac{188 \sigma_d}{\sqrt{\epsilon_{rd}}} \text{ nepers/m} \quad \text{..... (6.42)}$$

Example 6.4 : A gold parallel strip line of width 20 mm contains polyethylene dielectric material between the strip lines having a dielectric constant of 2.25 and thickness of dielectric as 4 mm. Determine the characteristic impedance and the phase velocity of the propagating microwave.

Solution : Given $w = 20$ mm, $d = 4$ mm, $\epsilon_{rd} = 2.5$.

for $w \gg d$, the characteristic impedance Z_0 is given by

$$Z_0 = \frac{377}{\sqrt{\epsilon_{rd}}} \left(\frac{d}{w} \right) \Omega$$

the phase velocity is given by

$$v_p = \frac{c}{\sqrt{\epsilon_{rd}}}$$

Example 6.4 : A gold parallel strip line of width 20 mm contains polyethylene dielectric material between the strip lines having a dielectric constant of 2.25 and thickness of dielectric as 4 mm. Determine the characteristic impedance and the phase velocity of the propagating microwave.

Solution : Given $w = 20$ mm, $d = 4$ mm, $\epsilon_{rd} = 2.5$.

From equation (6.37) for $w \gg d$, the characteristic impedance Z_0 is given by

$$Z_0 = \frac{377}{\sqrt{\epsilon_{rd}}} \left(\frac{d}{w} \right) \Omega = \frac{377}{\sqrt{2.25}} \left(\frac{4 \text{ mm}}{20 \text{ mm}} \right)$$

$$\therefore Z_0 = 50.27 \Omega$$

From equation (6.38), the phase velocity is given by

$$v_p = \frac{c}{\sqrt{\epsilon_{rd}}} = \frac{3 \times 10^8}{\sqrt{2.25}}$$

$$\therefore v_p = 2 \times 10^8 \text{ m/sec}$$

Example 6.5 : A lossless parallel strip line has copper conducting strips each of width $w = 18$ mm separated by quartz dielectric of dielectric constant 3.8 having thickness of 2.5 mm. The conductivity of copper is $5.8 \times 10^7 \text{ } \Omega/\text{m}$ and that of quartz is $2 \times 10^{-4} \text{ } \Omega/\text{m}$. The frequency of operation is 12 GHz.

Determine

- (a) Characteristic impedance of parallel strip line
- (b) Phase velocity of the propagating wave
- (c) Strip-line inductance
- (d) Strip-line capacitance
- (e) Series resistance for both strips
- (f) Shunt conductance of the dielectric
- (g) Attenuation constant for conductor losses
- (h) Attenuation constant for dielectric losses

Solution : Given $w = 18 \text{ mm}$, $\epsilon_{rd} = 3.8$, $d = 2.5 \text{ mm}$, $f = 12 \text{ GHz}$, $\sigma_c = 5.8 \times 10^7 \text{ } \Omega/\text{m}$,
 $\sigma_d = 2 \times 10^{-4} \text{ } \Omega/\text{m}$.

(a) Since $w \gg d$,

$$Z_0 = \frac{377}{\sqrt{\epsilon_{rd}}} \left(\frac{d}{w} \right)$$

(b) the phase velocity is given by

$$v_p = \frac{c}{\sqrt{\epsilon_{rd}}}$$

Solution : Given $w = 18 \text{ mm}$, $\epsilon_{rd} = 3.8$, $d = 2.5 \text{ mm}$, $f = 12 \text{ GHz}$, $\sigma_c = 5.8 \times 10^7 \text{ } \Omega/\text{m}$, $\sigma_d = 2 \times 10^{-4} \text{ } \Omega/\text{m}$.

(a) Since $w \gg d$, equation (6.37) for Z_0 can be used

$$\begin{aligned} \therefore Z_0 &= \frac{377}{\sqrt{\epsilon_{rd}}} \left(\frac{d}{w} \right) \\ &= \frac{377}{\sqrt{3.8}} \left(\frac{2.5 \text{ mm}}{18 \text{ mm}} \right) \end{aligned}$$

$$\therefore Z_0 = 26.86 \text{ } \Omega$$

(b) From equation (6.38), the phase velocity is given by

$$v_p = \frac{c}{\sqrt{\epsilon_{rd}}} = \frac{3 \times 10^8}{\sqrt{3.8}}$$

$$\therefore v_p = 1.54 \times 10^8 \text{ m/sec}$$

(c) The strip line inductance is given by

$$L = \frac{\mu_c d}{w} = \frac{\mu_0 \mu_{rc} d}{w}$$

(d) the strip line capacitance is given by

$$C = \frac{\epsilon_d w}{d} = \frac{\epsilon_0 \epsilon_{rd} w}{d}$$

(c) The strip line inductance is given by equation (6.33) as

$$L = \frac{\mu_c d}{w} = \frac{\mu_0 \mu_{rc} d}{w}$$

$$= \frac{4\pi \times 10^{-7} \times 1 \times 2.5 \times 10^{-3}}{18 \times 10^{-3}}$$

$$\therefore L = 0.175 \mu\text{H/m}$$

(d) From equation (6.34), the strip line capacitance is given by

$$C = \frac{\epsilon_d w}{d} = \frac{\epsilon_0 \epsilon_{rd} w}{d}$$

$$= \frac{8.854 \times 10^{-12} \times 3.8 \times 18 \times 10^{-3}}{2.5 \times 10^{-3}}$$

$$\therefore C = 242.25 \text{ pF/m}$$

(e) The series resistance for both strips is given by

$$R = \frac{2}{w} \sqrt{\frac{\pi f \mu_c}{\sigma_c}}$$
$$= \frac{2}{w} \sqrt{\frac{\pi f \mu_0 \mu_{rc}}{\sigma_c}}$$

(f) The shunt conductance of the dielectric substrate is given by

$$G = \frac{\sigma_d w}{d}$$

(e) The series resistance for both strips is given by equation (6.35) as

$$\begin{aligned}
 R &= \frac{2}{w} \sqrt{\frac{\pi f \mu_c}{\sigma_c}} \\
 &= \frac{2}{w} \sqrt{\frac{\pi f \mu_0 \mu_{rc}}{\sigma_c}} \\
 &= \frac{2}{18 \times 10^{-3}} \sqrt{\frac{\pi \times 12 \times 10^9 \times 4\pi \times 10^{-7} \times 1}{5.8 \times 10^7}} \\
 \therefore R &= 3.176 \Omega/\text{m}
 \end{aligned}$$

(f) The shunt conductance of the dielectric substrate is given by equation (6.36) as

$$\begin{aligned}
 G &= \frac{\sigma_d w}{d} \\
 &= \frac{2 \times 10^{-4} \times 18 \times 10^{-3}}{2.5 \times 10^{-3}} \\
 \therefore G &= 1.44 \times 10^{-3} \text{ S/m}
 \end{aligned}$$

(g) Attenuation constant for conductor losses is given by

$$\alpha_c = \frac{1}{d} \sqrt{\frac{\pi f \epsilon_d}{\sigma_c}}$$

(h) Attenuation constant for dielectric losses is given by

$$\alpha_d \approx \frac{188 \sigma_d}{\sqrt{\epsilon_{rd}}}$$

(g) Attenuation constant for conductor losses is given by equation (6.41) as

$$\alpha_c = \frac{1}{d} \sqrt{\frac{\pi f \epsilon_d}{\sigma_c}}$$

$$= \frac{1}{2.5 \times 10^{-3}} \sqrt{\frac{\pi \times 12 \times 10^9 \times 8.854 \times 10^{-12} \times 3.8}{5.8 \times 10^7}}$$

$$\therefore \alpha_c = 0.06 \text{ nepers/m}$$

(h) Attenuation constant for dielectric losses is given by equation (6.42) as

$$\alpha_d \simeq \frac{188 \sigma_d}{\sqrt{\epsilon_{rd}}}$$

$$\simeq \frac{(188) (2 \times 10^{-4})}{\sqrt{3.8}}$$

$$\alpha_d = 0.02 \text{ nepers/m}$$

Module 3: Chapter 2

Antenna Basics:

- Introduction
- Basic Antenna Parameters
- Patterns
- Beam Area
- Radiation Intensity
- Beam Efficiency
- Directivity and Gain
- Antenna Apertures
- Effective
- Height
- Bandwidth
- Radio Communication Link
- Antenna Field Zones & Polarization

Suggested Reading

Text 3: “Antennas and Wave Propagation”,
John D. Krauss, Ronald J Marhefka and Ahmad S Khan, 4th
Special Indian Edition , McGraw- Hill Education Pvt. Ltd., 2010.

Text 3:

Sections: 2.1- 2.11, 2.13, 2.15



Introduction :



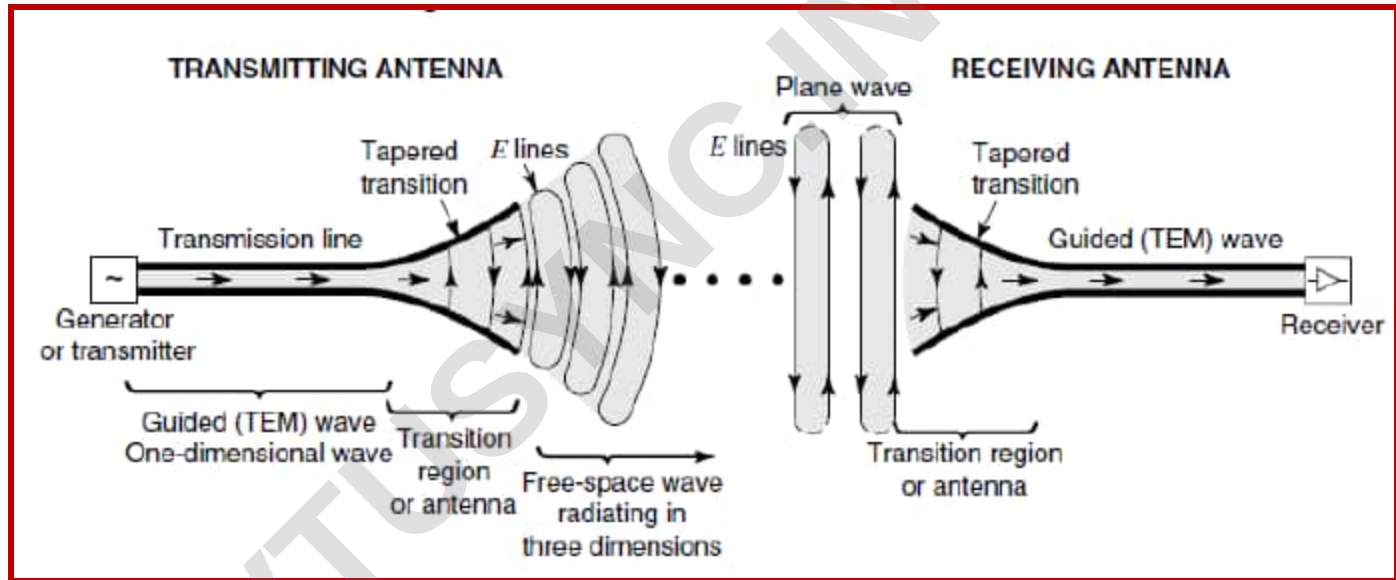
Definitions:

- An antenna is a metallic device (as a rod or wire) for radiating or receiving radio waves.
 - ✓ By Webster's Dictionary
- An antenna is a means for radiating or receiving radio waves.
 - ✓ IEEE Standard Definition (IEEE Std 145–1983)
- An antenna is the transitional structure between free-space and a guiding device.
 - ✓ The guiding device or transmission line may take the form of a coaxial line or a hollow pipe (waveguide), and it is used to transport electromagnetic energy from the transmitting source to the antenna or from the antenna to the receiver.
 - ✓ In the former case, we have a transmitting antenna and in the latter a receiving antenna.
- An antenna is basically a transducer. It converts radio frequency (RF) signal into an electromagnetic (EM) wave of the same frequency.

Definition of antenna

An antenna can be defined in the following different ways:

1. An antenna may be a piece of conducting material in the form of a wire, rod or any other shape with excitation.
2. An antenna is a source or radiator of electromagnetic waves.
3. An antenna is a sensor of electromagnetic waves.
4. An antenna is a transducer.
5. An antenna is an impedance matching device.
6. An antenna is a coupler between a generator and space or vice-versa.



The two-wire transmission line in Fig. 2-1a is connected to a radio-frequency generator (or transmitter). Along the uniform part of the line, energy is guided as a plane Transverse ElectroMagnetic Mode (TEM) wave with little loss. The spacing between wires is assumed to be a small fraction of a wavelength. Further on, the transmission line opens out in a tapered transition. As the separation approaches the order of a wavelength or more, the wave tends to be radiated so that the opened-out line acts like an antenna which launches a free-space wave. The currents on the transmission line flow out on the antenna and end there, but the fields associated with them keep on going.

From the circuit point of view, the antennas appear to the transmission lines as a resistance R_r , called the *radiation resistance*. It is not related to any resistance in the antenna itself but is a resistance coupled from space to the antenna terminals.

In the transmitting case, the radiated power is absorbed by objects at a distance: trees, buildings, the ground, the sky, and other antennas. In the receiving case, passive radiation from distant objects or active radiation from other antennas raises the apparent temperature of R_r . For lossless antennas this temperature has nothing

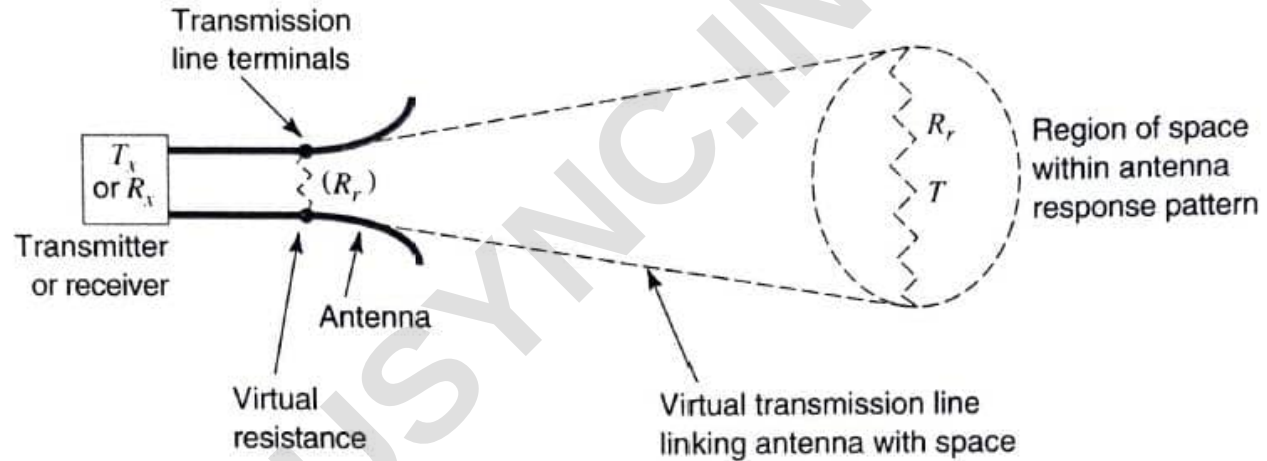


Figure 2-2 Schematic representation of region of space at temperature T linked via a virtual transmission line to an antenna.

I. Radiation Pattern

Radiation Pattern

Practically any antenna can not radiate energy with same strength uniformly, in all directions. It is found that the radiation is large in one direction while zero or minimum in other directions. The radiation from the antenna in any direction is measured in terms of **field strength** at a point located at a particular distance from an antenna. The field strength can be calculated by measuring voltages at two points on an electric lines of force and then dividing by distance between two points. Hence unit of the radiation pattern is volt per meter. Generally the field strength is expressed in millivolts per meter. The radiation pattern of an antenna is the important characteristic of antenna because it indicates the distribution of the energy radiated by an antenna in the space.

In general, the radiation pattern is nothing but a graph which shows the variation of actual field strength of electromagnetic field at all the points equidistant from the antenna. Hence it is a three dimensional graph. There are two basic radiation patterns. If the radiation of the antenna is represented graphically as a function of direction it is called **radiation pattern**. But if the radiation of the antenna is expressed in terms of the field strength E (in V/m), then the graphical representation is called **Field Strength Pattern** or **Field Radiation Pattern**. Similarly if the radiation of the antenna is expressed in terms of the power per unit solid angle, then the graphical representation is called **Power Radiation Pattern** or simply **Power Pattern**.

To completely specify the radiation pattern with respect to field intensity and polarization requires three patterns:

1. The θ component of the electric field as a function of the angles θ and ϕ or $E_\theta(\theta, \phi)$ ($V\ m^{-1}$) as in Figs. 2-3 and 2-4.
2. The ϕ component of the electric field as a function of the angles θ and ϕ or $E_\phi(\theta, \phi)$ ($V\ m^{-1}$).
3. The phases of these fields as a function of the angles θ and ϕ or $\delta_\theta(\theta, \phi)$ and $\delta_\phi(\theta, \phi)$ (rad or deg).

Figure 2-3

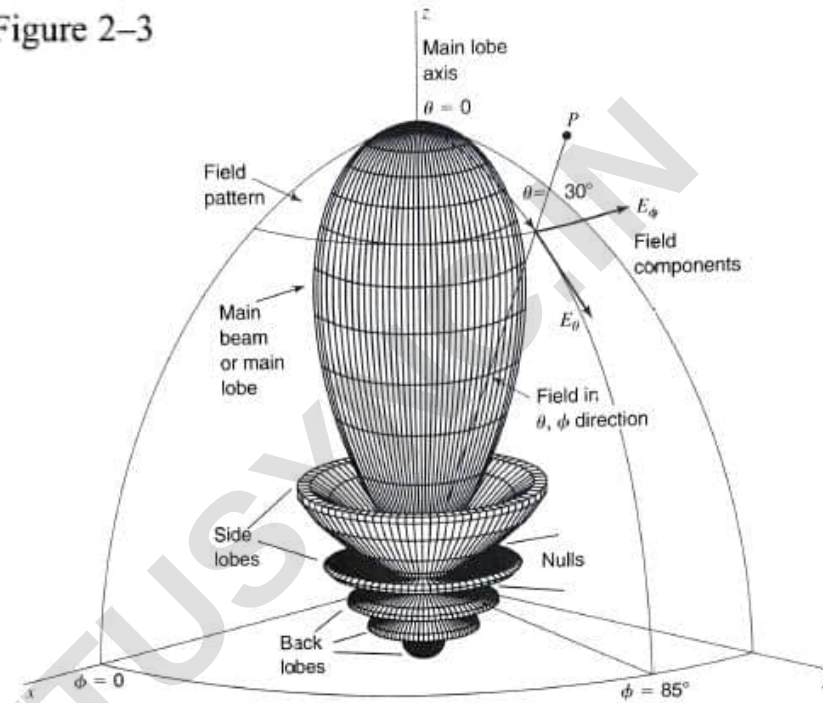
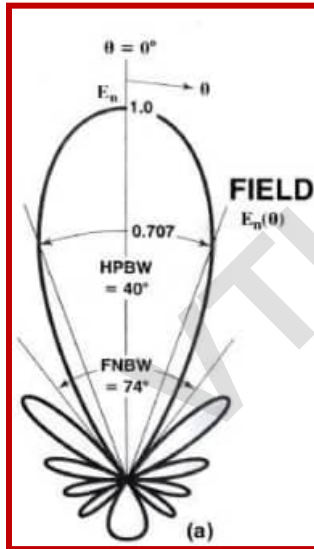


Figure 2-3 shows a three-dimensional field pattern with pattern radius r (from origin to pattern boundary at the dot) proportional to the field intensity in the direction θ and ϕ . The pattern has its *main lobe* (maximum radiation) in the z direction ($\theta = 0$) with *minor lobes* (side and back) in other directions.

Normalized field pattern

In general, the field pattern is expressed mathematically in terms of **relative field pattern** which is commonly called **normalized field pattern**. The **normalized field pattern** is defined as the ratio of the field component to its maximum value. Basically normalized field pattern is a dimensionless quantity with **maximum value equal to one (unity)**.

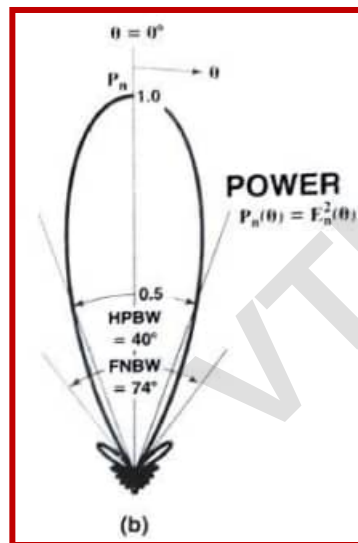


$$\text{Normalized field pattern} = E_\theta(\theta, \phi)_n = \frac{E_\theta(\theta, \phi)}{E_\theta(\theta, \phi)_{\max}} \quad (\text{dimensionless})$$

Normalized Power pattern

When the radiation in a given direction is expressed in terms of power per unit solid angle, the pattern is called **power radiation pattern**.

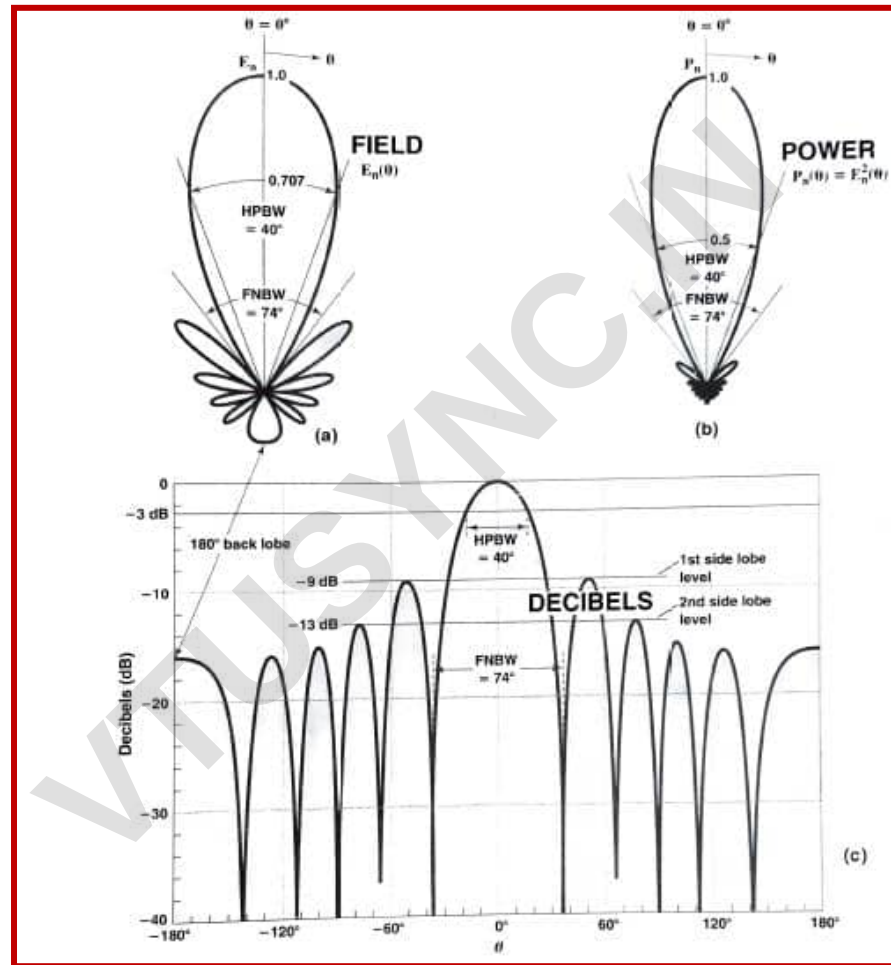
Normalizing this power with respect to its maximum value yields a *normalized power pattern* as a function of angle which is a dimensionless number with a maximum value of unity. Thus, the normalized power pattern (Fig. 2-4b) is given by



$$\text{Normalized power pattern} = P_n(\theta, \phi)_n = \frac{S(\theta, \phi)}{S(\theta, \phi)_{\max}} \quad (\text{dimensionless})$$

where $S(\theta, \phi)$ = Poynting vector = $[E_\theta^2(\theta, \phi) + E_\phi^2(\theta, \phi)]/Z_0$, W m^{-2}
 $S(\theta, \phi)_{\max}$ = maximum value of $S(\theta, \phi)$, W m^{-2}
 Z_0 intrinsic impedance of space = 376.7Ω

The decibel level is given by $\text{dB} = 10 \log_{10} P_n(\theta, \phi)$



Although the radiation pattern characteristics of an antenna involve three-dimensional vector fields for a full representation, several simple single-valued scalar quantities can provide the information required for many engineering applications. These are:

- Half-power beamwidth, HPBW
- Beam area, Ω_A
- Beam efficiency, ε_M
- Directivity D or gain G
- Effective aperture A_e

2. Beam width:

It is the measure of sharpness of beam

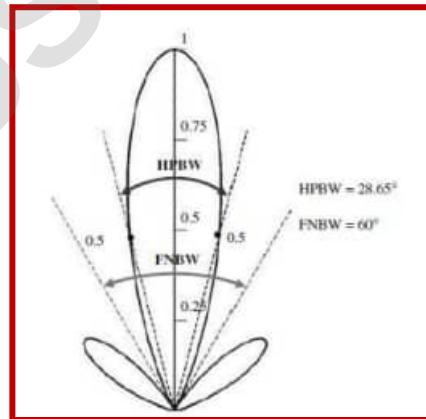
Half Power Beam width (HPBW) :

“The Half Power Beam width (HPBW) is the angular separation in which the magnitude of the radiation pattern decrease by 50% (or -3 dB) from the peak of the main beam.”

First Null Beam Width (FNBW):

“The angular span between the first pattern nulls adjacent to the main lobe, is called as the First Null Beam Width.”

The half-power level occurs at those angles θ and ϕ for which $E_{\theta}(\theta, \phi)_n = 1/\sqrt{2} = 0.707$.



EXAMPLE 2-3.1 Half-Power Beamwidth

An antenna has a field pattern given by

$$E(\theta) = \cos^2 \theta \quad \text{for } 0^\circ \leq \theta \leq 90^\circ$$

Find the half-power beamwidth (HPBW).

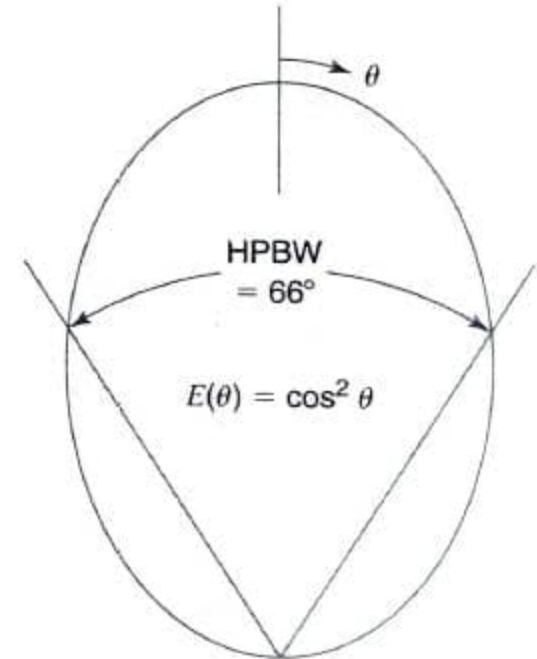
■ Solution

$E(\theta)$ at half power = 0.707.

Thus $0.707 = \cos^2 \theta$ so

$\cos \theta = \sqrt{0.707}$ and $\theta = 33^\circ$

$$\text{HPBW} = 2\theta = 66^\circ \quad \text{Ans.}$$



EXAMPLE 2-3.2 Half-Power Beamwidth and First Null Beamwidth

An antenna has a field pattern given by $E(\theta) = \cos \theta \cos 2\theta$ for $0^\circ \leq \theta \leq 90^\circ$. Find (a) the half-power beamwidth (HPBW) and (b) the beamwidth between first nulls (FNBW).

■ Solution

(a) $E(\theta)$ at half power = 0.707. Thus $0.707 = \cos \theta \cos 2\theta = 1/\sqrt{2}$.

$$\cos 2\theta = \frac{1}{\sqrt{2} \cos \theta} \quad 2\theta = \cos^{-1} \left(\frac{1}{\sqrt{2} \cos \theta} \right) \quad \text{and}$$

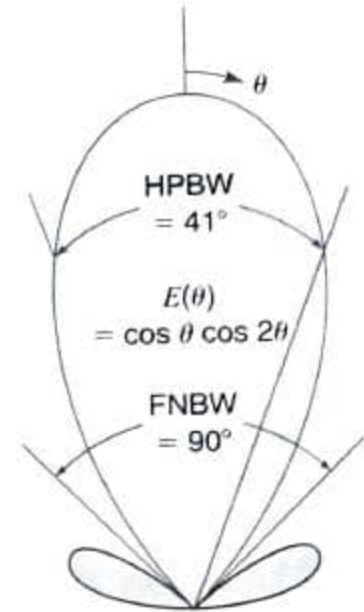
$$\theta = \frac{1}{2} \cos^{-1} \left(\frac{1}{\sqrt{2} \cos \theta'} \right)$$

Iterating with $\theta' = 0$ as a first guess, $\theta = 22.5^\circ$. Setting $\theta' = 22.5^\circ$, $\theta = 20.03^\circ$, etc., until after next iteration $\theta = \theta' = 20.47^\circ \cong 20.5^\circ$ and

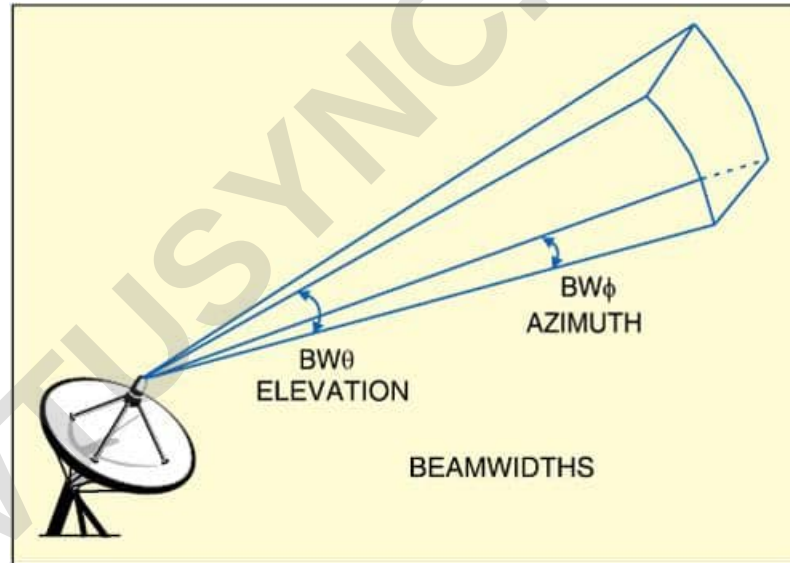
$$\text{HPBW} = 2\theta = 41^\circ \quad \text{Ans. (a)}$$

(b) $0 = \cos \theta \cos 2\theta$, so $\theta = 45^\circ$ and

$$\text{FNBW} = 2\theta = 90^\circ \quad \text{Ans. (b)}$$



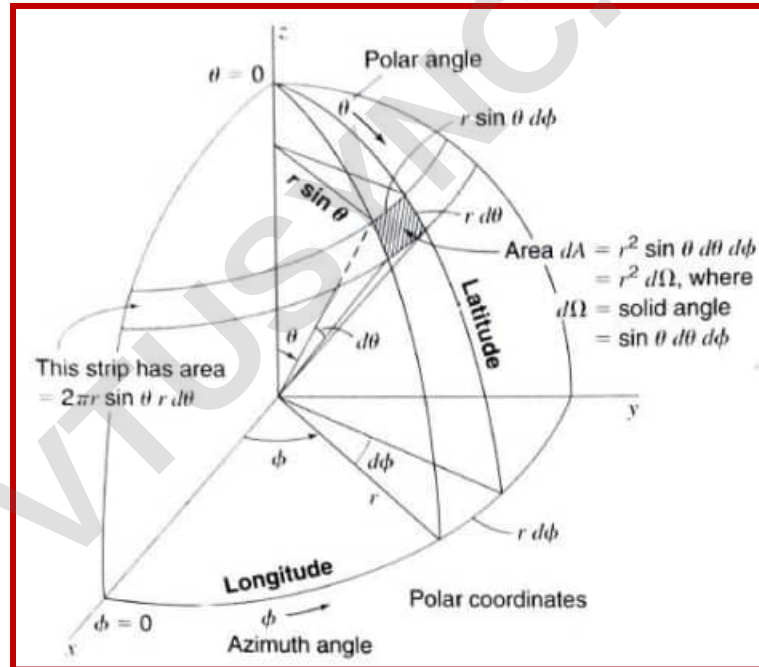
3. Beam Area or Beam Solid Angle

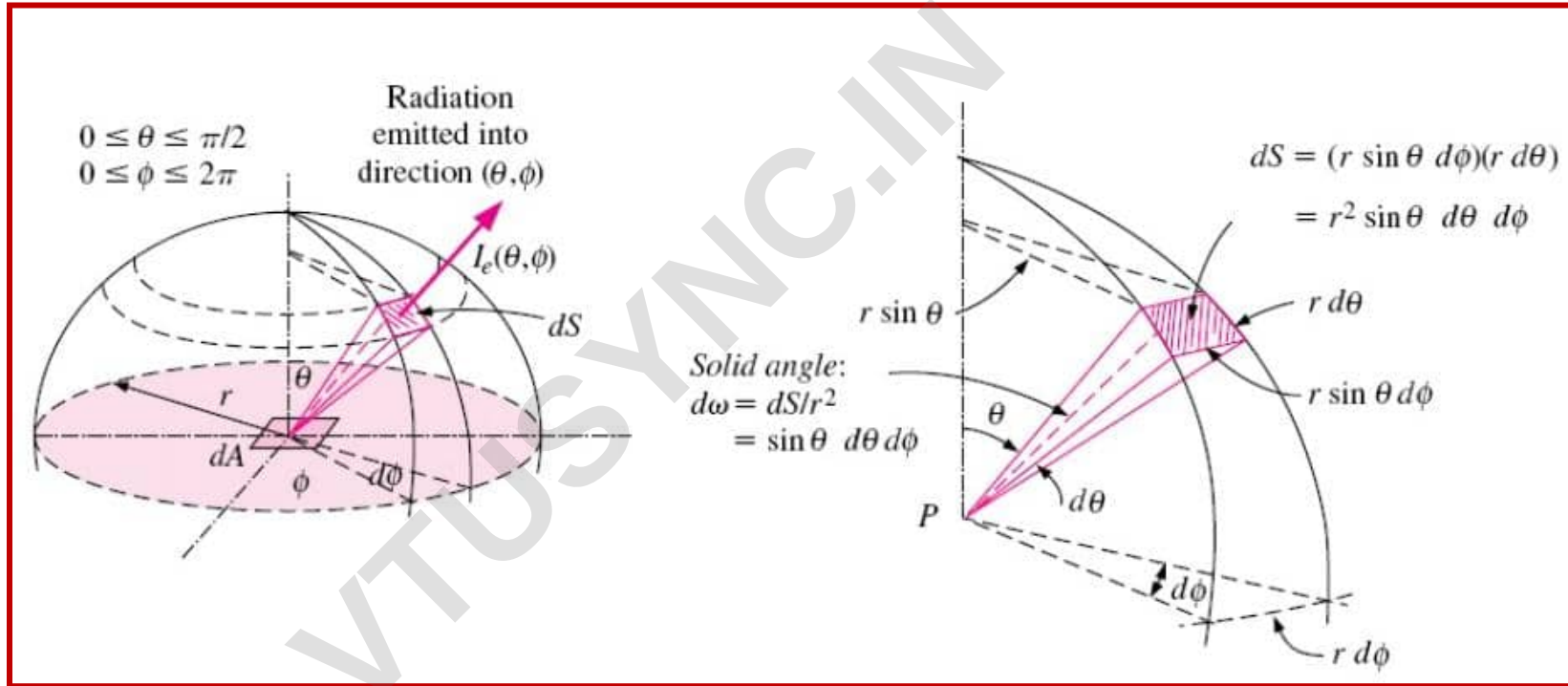


Beam Area (or Beam Solid Angle) Ω_A

In polar two-dimensional coordinates an incremental area dA on the surface of a sphere is the product of the length $r d\theta$ in the θ direction (latitude) and $r \sin \theta d\phi$ in the ϕ direction (longitude), as shown in Fig. 2-5.

$$\text{Thus, } dA = (r d\theta)(r \sin \theta d\phi) = r^2 d\Omega$$





Thus,

$$dA = (r d\theta)(r \sin \theta d\phi) = r^2 d\Omega \quad (1)$$

where

$d\Omega$ = solid angle expressed in steradians (sr) or square degrees($^\square$)

$d\Omega$ = solid angle subtended by the area dA

The area of the strip of width $r d\theta$ extending around the sphere at a constant angle θ is given by $(2\pi r \sin \theta)(r d\theta)$. Integrating this for θ values from 0 to π yields the area of the sphere. Thus,

$$\text{Area of sphere} = 2\pi r^2 \int_0^\pi \sin \theta d\theta = 2\pi r^2 [-\cos \theta]_0^\pi = 4\pi r^2 \quad (2)$$

where 4π = solid angle subtended by a sphere, sr

Thus,

$$\begin{aligned} 1 \text{ steradian} &= 1 \text{ sr} = (\text{solid angle of sphere})/(4\pi) \\ &= 1 \text{ rad}^2 = \left(\frac{180}{\pi}\right)^2 (\text{deg}^2) = 3282.8064 \text{ square degrees} \end{aligned} \quad (3)$$

Therefore,

$$\begin{aligned} 4\pi \text{ steradians} &= 3282.8064 \times 4\pi = 41,252.96 \cong 41,253 \text{ square degrees} = 41,253^\square \\ &= \text{solid angle in a sphere} \end{aligned} \quad (4)$$

- Beam Area in terms of Solid Angle
- Beam Area in terms of Normalized Power Pattern
- Beam Area in terms of Half Power Points

The beam area or *beam solid angle* or Ω_A of an antenna (Fig. 2-5b) is given by the integral of the normalized power pattern over a sphere (4π sr)

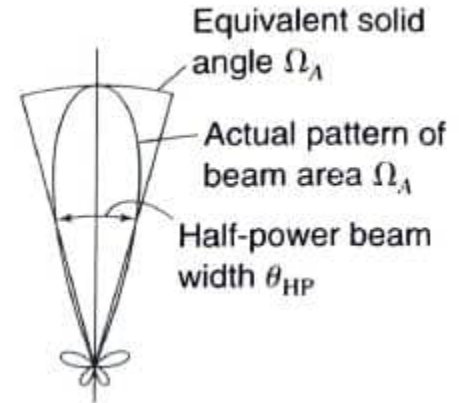
$$\Omega_A = \int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} P_n(\theta, \phi) \sin \theta d\theta d\phi \quad (5a)$$

and

$$\Omega_A = \int \int_{4\pi} P_n(\theta, \phi) d\Omega \quad (\text{sr}) \quad \text{Beam area} \quad (5b)$$

where $d\Omega = \sin \theta d\theta d\phi$, sr.

The beam area Ω_A is the solid angle through which all of the power radiated by the antenna would stream if $P(\theta, \phi)$ maintained its maximum value over Ω_A and was zero elsewhere. Thus the power radiated = $P(\theta, \phi)\Omega_A$ watts.



The *beam area* of an antenna can often be described *approximately* in terms of the angles subtended by the *half-power points* of the main lobe in the two principal planes. Thus,

$$\text{Beam area} \cong \Omega_A \cong \theta_{\text{HP}} \phi_{\text{HP}} \quad (\text{sr}) \quad (6)$$

where θ_{HP} and ϕ_{HP} are the *half-power beamwidths* (HPBW) in the two principal planes, minor lobes being neglected.

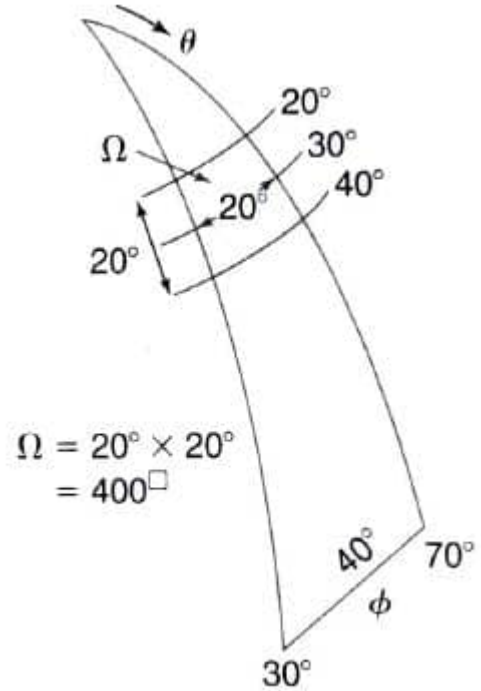
EXAMPLE 2-4.1 Solid Angle of Area in Square Degrees

Find the number of square degrees in the solid angle Ω on a spherical surface that is between $\theta = 20^\circ$ and $\theta = 40^\circ$ (or 70° and 50° north latitude) and between $\phi = 30^\circ$ and $\phi = 70^\circ$ (30° and 70° east longitude).

■ Solution

From (1)

$$\Omega = \int_{30^\circ}^{70^\circ} d\phi \int_{20^\circ}^{40^\circ} \sin \theta d\theta$$



■ Solution

From (1)

$$\begin{aligned}\Omega &= \int_{30^\circ}^{70^\circ} d\phi \int_{20^\circ}^{40^\circ} \sin \theta d\theta \\ &= \frac{40}{360} 2\pi [-\cos \theta]_{20}^{40} \\ &= 0.222\pi \times 0.173 = 0.121 \text{ steradians (sr)} \\ &= 0.121 \times 3283 = 397 \text{ square degrees} = 397^\square\end{aligned}$$

EXAMPLE 2-4.2 Beam Area Ω_A of Antenna with $\cos^2\theta$ Pattern

An antenna has a field pattern given by $E(\theta) = \cos^2\theta$ for $0^\circ \leq \theta \leq 90^\circ$. This is the same pattern of Example 2-3.1. Find the beam area of this pattern.

■ **Solution**

From (5)

$$\Omega_A = \int_0^{2\pi} \int_0^\pi \cos^4\theta \sin\theta \, d\theta \, d\phi$$

$$\begin{aligned}\Omega_A &= \int_0^{2\pi} \int_0^{\pi} \cos^4 \theta \sin \theta \, d\theta \, d\phi \\ &= \int_0^{2\pi} d\phi \int_0^{\pi} \cos^4 \theta \sin \theta \, d\theta \\ &= [2\pi - 0] \int_0^{\pi} \cos^4 \theta \sin \theta \, d\theta\end{aligned}$$

Let $u = \cos \theta$

$$du = -\sin \theta \, d\theta$$

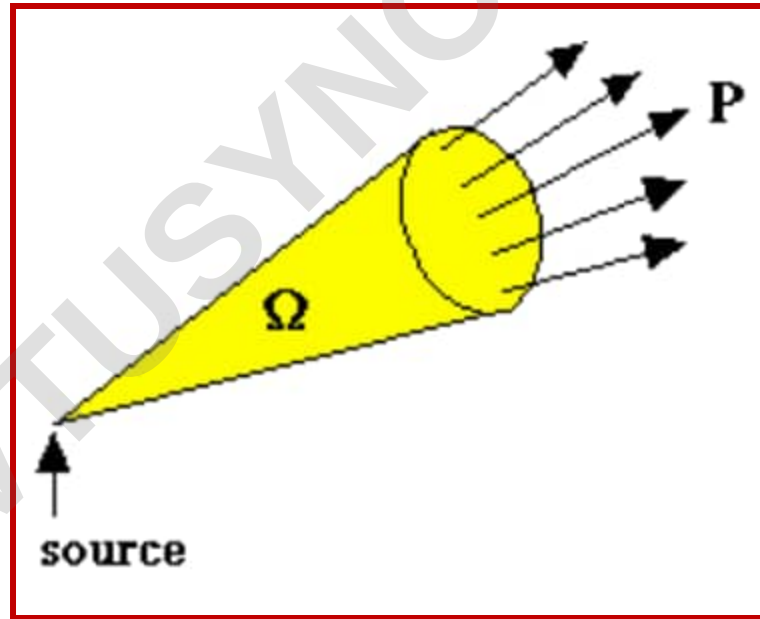
$$-du = \sin \theta \, d\theta$$

$$\begin{aligned}&\int_0^{\pi} u^4 (-du) \\ &= -\int_0^{\pi} u^4 \, du = -\left[\frac{u^5}{5} + C \right]_0^{\pi} = \left[\frac{\cos^5 \theta}{5} \right]_0^{\pi} \\ &= \frac{1}{5} [\cos^5 \pi - \cos^5 0] = \frac{1}{5}\end{aligned}$$

$$= 2\pi \times \frac{1}{5} = 1.26 \text{ sr}$$

$$\begin{aligned}\Omega_A &= \int_0^{2\pi} \int_0^{\pi} \cos^4 \theta \sin \theta \, d\theta \, d\phi \\ &= \frac{2\pi}{5} = 1.26 \text{ sr} \quad \text{Ans.}\end{aligned}$$

4. Radiation Intensity



Radiation Intensity

The power radiated from an antenna per unit solid angle is called the *radiation intensity* U (watts per steradian or per square degree). The normalized power pattern of the previous section can also be expressed in terms of this parameter as the ratio of the radiation intensity $U(\theta, \phi)$, as a function of angle, to its maximum value. Thus,

$$P_n(\theta, \phi) = \frac{U(\theta, \phi)}{U(\theta, \phi)_{\max}} = \frac{S(\theta, \phi)}{S(\theta, \phi)_{\max}} \quad (1)$$

5. Beam Efficiency

Beam Efficiency

The (total) *beam area* Ω_A (or *beam solid angle*) consists of the main beam area (or solid angle) Ω_M plus the minor-lobe area (or solid angle) Ω_m .¹ Thus,

$$\Omega_A = \Omega_M + \Omega_m \quad (1)$$

The ratio of the main beam area to the (total) beam area is called the (main) *beam efficiency* ε_M . Thus,

$$\text{Beam efficiency} = \varepsilon_M = \frac{\Omega_M}{\Omega_A} \quad (\text{dimensionless}) \quad (2)$$

The ratio of the minor-lobe area (Ω_m) to the (total) beam area is called the *stray factor*. Thus,

$$\varepsilon_m = \frac{\Omega_m}{\Omega_A} = \text{stray factor} \quad (3)$$

It follows that

$$\varepsilon_M + \varepsilon_m = 1 \quad (4)$$

6. Directivity and Gain

Directivity D and Gain G

The directivity D and the gain G are probably the most important parameters of an antenna.

The *directivity* of an antenna is equal to the ratio of the maximum power density $P(\theta, \phi)_{\max}$ (watts/m²) to its average value over a sphere as observed in the far field of an antenna.

$$D = \frac{P(\theta, \phi)_{\max}}{P(\theta, \phi)_{\text{av}}} \quad \text{Directivity from pattern}$$

(1)

The directivity is a dimensionless ratio ≥ 1 .

The average power density over a sphere is given by

$$P(\theta, \phi)_{av} = \frac{1}{4\pi} \int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} P(\theta, \phi) \sin \theta d\theta d\phi$$

$$= \frac{1}{4\pi} \iint_{4\pi} P(\theta, \phi) d\Omega \quad (\text{W sr}^{-1})$$

Therefore, the directivity

$$D = \frac{P(\theta, \phi)_{\max}}{(1/4\pi) \iint_{4\pi} P(\theta, \phi) d\Omega} = \frac{1}{(1/4\pi) \iint_{4\pi} [P(\theta, \phi)/P(\theta, \phi)_{\max}] d\Omega}$$

and

$$D = \frac{4\pi}{\iint_{4\pi} P_n(\theta, \phi) d\Omega} = \frac{4\pi}{\Omega_A} \quad \text{Directivity from beam area } \Omega_A$$

where $P_n(\theta, \phi) d\Omega = P(\theta, \phi)/P(\theta, \phi)_{\max} = \text{normalized power pattern}$

The smaller the beam area, the larger the directivity D . For an antenna that radiates over only half a sphere the beam area $\Omega_A = 2\pi$ sr (Fig. 2-6) and the directivity is

$$D = \frac{4\pi}{2\pi} = 2 \quad (= 3.01 \text{ dBi}) \quad (5)$$

where dBi = decibels over isotropic

Note that the idealized *isotropic antenna* ($\Omega_A = 4\pi$ sr) has the lowest possible directivity $D = 1$. All actual antennas have directivities greater than 1 ($D > 1$). The simple short dipole has a beam area $\Omega_A = 2.67\pi$ sr and a directivity $D = 1.5$ ($= 1.76$ dBi).

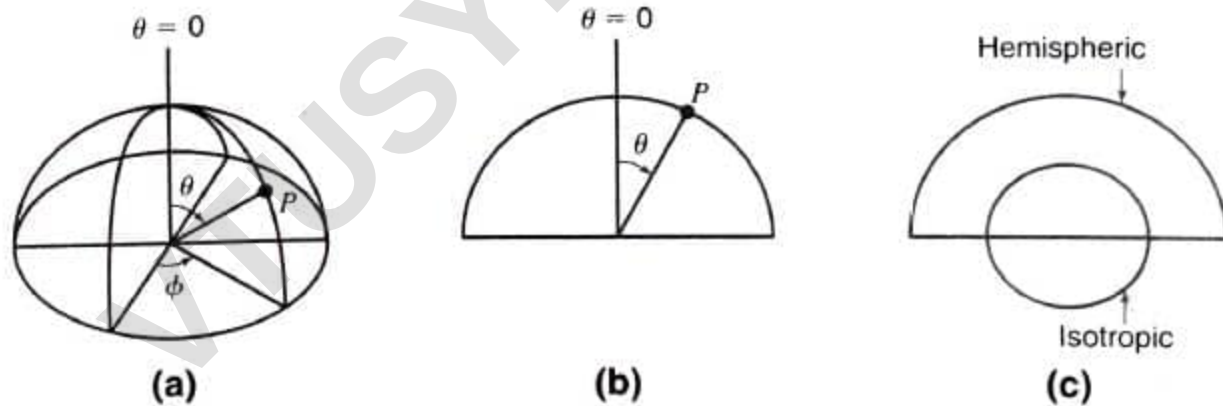


Figure 2-6 Hemispheric power patterns, (a) and (b), and comparison with isotropic pattern (c).

The gain G of an antenna is an actual or realized quantity which is less than the directivity D due to ohmic losses in the antenna or its radome (if it is enclosed). In transmitting, these losses involve power fed to the antenna which is not radiated but heats the antenna structure. A mismatch in feeding the antenna can also reduce the gain. The ratio of the gain to the directivity is the *antenna efficiency factor*. Thus,

$$G = kD \quad (6)$$

where k = efficiency factor ($0 \leq k \leq 1$), dimensionless.

In many well-designed antennas, k may be close to unity. In practice, G is always less than D , with D its maximum idealized value.

Gain can be measured by comparing the maximum power density of the Antenna Under Test (AUT) with a reference antenna of known gain, such as a short dipole. Thus,

$$\text{Gain} = G = \frac{P_{\max}(\text{AUT})}{P_{\max}(\text{ref. ant.})} \times G(\text{ref. ant.}) \quad (7)$$

If the half-power beamwidths of an antenna are known, its directivity

$$D = \frac{41,253^\circ}{\theta_{HP}^\circ \phi_{HP}^\circ} \quad (8)$$

where

$41,253^\circ$ = number of square degrees in sphere = $4\pi(180/n)^2$ square degrees ($^\circ$)

θ_{HP}° = half – power beamwidth in one principal plane

ϕ_{HP}° = half – power beamwidth in other principal plane

Since (8) neglects minor lobes, a better approximation is a

$$D = \frac{40,000^\circ}{\theta_{HP}^\circ \phi_{HP}^\circ} \quad \text{Approximate directivity} \quad (9)$$

If the antenna has a main half-power beamwidth (HPBW) = 20° in both principal planes, its directivity

$$D = \frac{40,000^\circ}{400^\circ} = 100 \text{ or } 20 \text{ dBi} \quad (10)$$

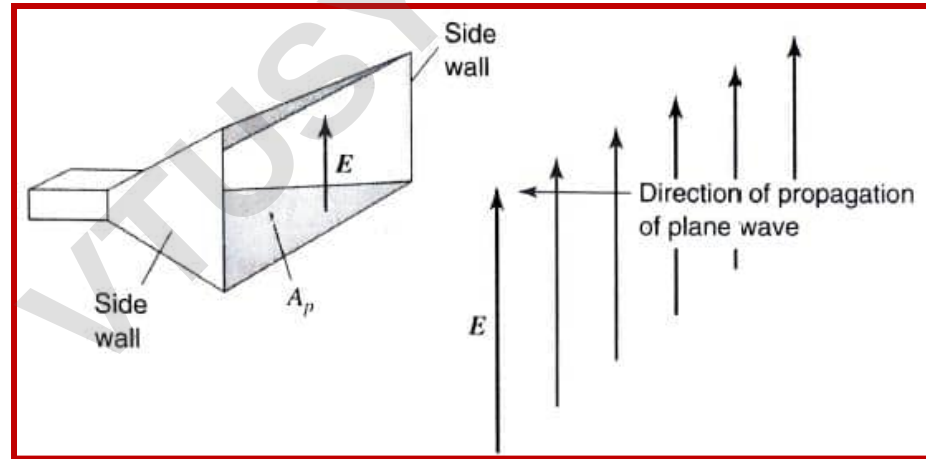
which means that the antenna radiates 100 times the power in the direction of the main beam as a nondirectional, isotropic antenna.

6. Antenna Apertures

Antenna Apertures

The concept of aperture is most simply introduced by considering a receiving antenna. Suppose that the receiving antenna is a rectangular electromagnetic horn immersed in the field of a uniform plane wave as suggested in Fig. 2-7. Let the Poynting vector, or power density, of the plane wave be S watts per square meter and the area, or physical aperture of the horn, be A_p square meters. If the horn extracts all the power from the wave over its entire physical aperture, then the total power P absorbed from the wave is

$$P = \frac{E^2}{Z} A_p = S A_p \quad (\text{W}) \quad (1)$$



Thus, the electromagnetic horn may be regarded as having an aperture, the total power it extracts from a passing wave being proportional to the aperture or area of its mouth.

But the field response of the horn is NOT uniform across the aperture A because E at the sidewalls must equal zero. Thus, the *effective aperture* A_e of the horn is less than the *physical aperture* A_p as given by

$$\varepsilon_{ap} = \frac{A_e}{A_p} \quad (\text{dimensionless}) \quad \text{Aperture efficiency} \quad (2)$$

where ε_{ap} = aperture efficiency.

Consider now an antenna with an *effective aperture* A_e , which radiates all of its power in a conical pattern of beam area Ω_A , as suggested in Fig. 2-8. Assuming a uniform field E_a over the aperture, the power radiated is

$$P = \frac{E_a^2}{Z_0} A_e \quad (\text{W}) \quad (3)$$

where Z_0 = intrinsic impedance of medium (377Ω for air or vacuum).

Assuming a uniform field E_r in the far field at a distance r , the power radiated is also given by

$$P = \frac{E_r^2}{Z_0} r^2 \Omega_A \quad (\text{W}) \quad (4)$$

Equating (3) and (4) and noting that $E_r = E_a A_e / r \lambda$ yields the aperture-beam-area relation

$$\lambda^2 = A_e \Omega_A \quad (\text{m}^2) \quad \text{Aperture-beam-area relation} \quad (5)$$

where Ω_A = beam area (sr).

Thus, if A_e is known, we can determine Ω_A (or vice versa) at a given wavelength. From (5) and (2-7-4) it follows that the directivity

$$D = 4\pi \frac{A_e}{\lambda^2} \quad \text{Directivity from aperture} \quad (6)$$

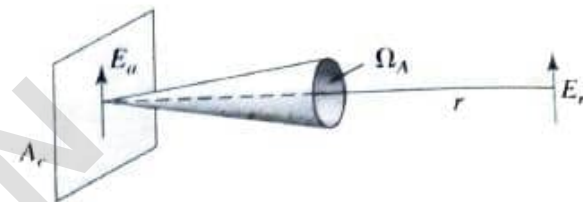


Figure 2-8 Radiation over beam area Ω_A from aperture A_e .

Three expressions have now been given for the *directivity D*. They are

$$D = \frac{P(\theta, \phi)_{\max}}{P(\theta, \phi)_{\text{av}}} \quad (\text{dimensionless}) \quad \text{Directivity from pattern} \quad (8)$$

$$D = \frac{4\pi}{\Omega_A} \quad (\text{dimensionless}) \quad \text{Directivity from pattern} \quad (9)$$

$$D = 4\pi \frac{A_e}{\lambda^2} \quad (\text{dimensionless}) \quad \text{Directivity from aperture} \quad (10)$$

7. Effective Height

Effective Height

The *effective height* h (meters) of an antenna is another parameter related to the aperture. Multiplying the effective height by the incident field E (volts per meter) of the same polarization gives the voltage V induced. Thus,

$$V = hE \quad (1)$$

Accordingly, the effective height may be defined as the ratio of the induced voltage to the incident field or

$$h = \frac{V}{E} \quad (\text{m}) \quad (2)$$

For an antenna of radiation resistance R_r matched to its load, the power delivered to the load is equal to

$$P = \frac{1}{4} \frac{V^2}{R_r} = \frac{h^2 E^2}{4 R_r} \quad (\text{W}) \quad (4)$$

In terms of the effective aperture the same power is given by

$$P = S A_e = \frac{E^2 A_e}{Z_0} \quad (\text{W}) \quad (5)$$

where $Z_0 =$ intrinsic impedance of space ($= 377 \Omega$)

Equating (4) and (5), we obtain

$$h_e = 2 \sqrt{\frac{R_r A_e}{Z_0}} \quad (\text{m}) \quad \text{and} \quad A_e = \frac{h_e^2 Z_0}{4 R_r} \quad (\text{m}^2) \quad (6)$$

Thus, effective height and effective aperture are related via radiation resistance and the intrinsic impedance of space.

The Radio Communication Link

The Radio Communication Link

The usefulness of the aperture concept is well illustrated by using it to derive the important Friis transmission formula published in 1946 by Harald T. Friis (1) of the Bell Telephone Laboratory.

Referring to Fig. 2-14, the formula gives the power received over a radio communication link. Assuming lossless, matched antennas, let the transmitter feed a power P_t to a transmitting antenna of effective aperture A_{et} . At a distance r a receiving antenna of effective aperture A_{er} intercepts some of the power radiated by the transmitting antenna and delivers it to the receiver R . Assuming for the moment that the transmitting antenna is isotropic, the power per unit area available at the receiving antenna is

$$S_r = \frac{P_t}{4\pi r^2} \quad (\text{W})$$

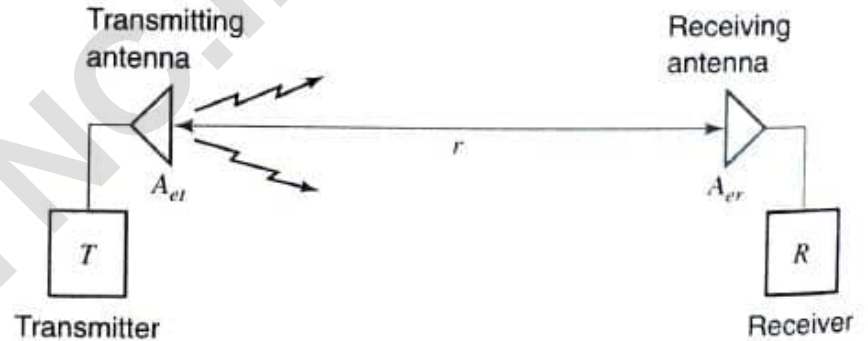


Figure 2-14 Communication circuit with waves from transmitting antenna arriving at the receiving antenna by a direct path of length r .

(1)

If the antenna has gain G_t , the power per unit area available at the receiving antenna will be increased in proportion as given by

$$S_r = \frac{P_t G_t}{4\pi r^2} \quad (\text{W}) \quad (2)$$

Now the power collected by the lossless, matched receiving antenna of effective aperture A_{er} is

$$P_r = S_r A_{er} = \frac{P_t G_t A_{er}}{4\pi r^2} \quad (\text{W}) \quad (3)$$

The gain of the transmitting antenna can be expressed as

$$G_t = \frac{4\pi A_{et}}{\lambda^2} \quad (4)$$

Substituting this in (3) yields the *Friis transmission formula*

$$\frac{P_r}{P_t} = \frac{A_{er} A_{et}}{r^2 \lambda^2} \quad (\text{dimensionless}) \quad \text{Friis transmission formula} \quad (5)$$

where

P_r = received power, W

P_t = transmitted power, W

A_{et} = effective aperture of transmitting antenna, m^2

A_{er} = effective aperture of receiving antenna, m^2

r = distance between antennas, m

λ = wavelength, m

EXAMPLE 2-11.1 Radio Communication Link

A radio link has a 15-W transmitter connected to an antenna of 2.5 m^2 effective aperture at 5 GHz. The receiving antenna has an effective aperture of 0.5 m^2 and is located at a 15-km line-of-sight distance from the transmitting antenna. Assuming lossless, matched antennas, find the power delivered to the receiver.

■ Solution

$$P = P_t \frac{A_{et} A_{er}}{r^2 \lambda^2} = 15 \frac{2.5 \times 0.5}{15^2 \times 10^6 \times 0.06^2} = 23 \mu\text{W} \quad \text{Ans.}$$

CBCS SCHEME

USN

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15EC71

Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019 Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing
ONE full question from each module.

Module-3

- 5 a. Explain the construction and field pattern for microstrip line. (06 Marks)
- b. Explain the following terms as related to antenna system : (06 Marks)
- i) directivity ii) beam efficiency iii) effective aperture.
- c. The effective apertures of transmitting and receiving antennas in a communication system are $8\lambda^2$ and $12\lambda^2$ respectively. With a separation of 1.5km between them. The EM wave travelling with frequency of 6 MHz and the total input power is 25KW. Find the power received by the receiving antenna. (04 Marks)

Friis Transmission Equation (Other equation)

Friis Transmission Formula can be used to study RF communication links. The formula can be used in situations where the **distance between two antennas are known and a suitable antenna** needs to be found.

Using Friis transmission equation, one can solve for the antenna gains needed at either the transmitter or receiver in order to meet certain design specification

$$\frac{P_{\text{receiver}}}{P_{\text{transmitter}}} = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2$$

Where P_{receiver} is the received power in Watts (W),

$P_{\text{transmitter}}$ is the transmitted power,

G_t is the transmitting antenna's gain,

G_r is the receiving antenna's gain,

λ is the wavelength of the transmitted and received signal in meters, and

R is the distance between the antennas in meters.

The gain of the antennas is not in decibels.

Problem:

If an engineer is designing a communications link with a receiver antenna gain of 10dB and transmitter antenna gain of 12dB at a frequency of 10GHz, how much power will be received at the receiver if the transmitter is transmitting with a power of 25 watts across a distance of 1 kilo meter.

$$\frac{P_{\text{receiver}}}{P_{\text{transmitter}}} = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2$$

CBGS SCHEME

USN

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15EC71

Seventh Semester B.E. Degree Examination, June/July 2019 Microwave and Antennas

Module-3

- 5 a. A lossless parallel strip line has a conducting strip width W . The substrate dielectric separating the two conducting strips has a relative dielectric constant ϵ_{rd} of 6 and a thickness d of 4mm. Calculate: i) The required width W of the conducting strip in order to have a characteristic impedance of 50Ω ; ii) The strip-line capacitance. (04 Marks)
- b. Discuss different types of losses in microstrip lines. (06 Marks)
- c. Calculate the exact directivity for 3 dimensional source having the pattern $U = U_m \sin^2 \theta \sin^2 \phi$ where $0 \leq \theta \leq \pi$, $0 \leq \phi \leq \pi$. (06 Marks)

OR

- 6 a. Show that maximum effective aperture of a $\lambda/2$ dipole antenna is $0.13\lambda^2$. (06 Marks)
- b. With the aid of schematic diagram explain coplanar strip line. (05 Marks)
- c. Compute the power received by receiving antenna kept at a distance of 100km by a transmitter radiating at 3MHz. Assume $G_T = 40$ and $G_R = 15$ and $P_T = 1000$ kW. Derive the relation used. (05 Marks)

Antenna Field Zones

CBCS SCHEME

USN

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15EC71

Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019 **Microwaves and Antennas**

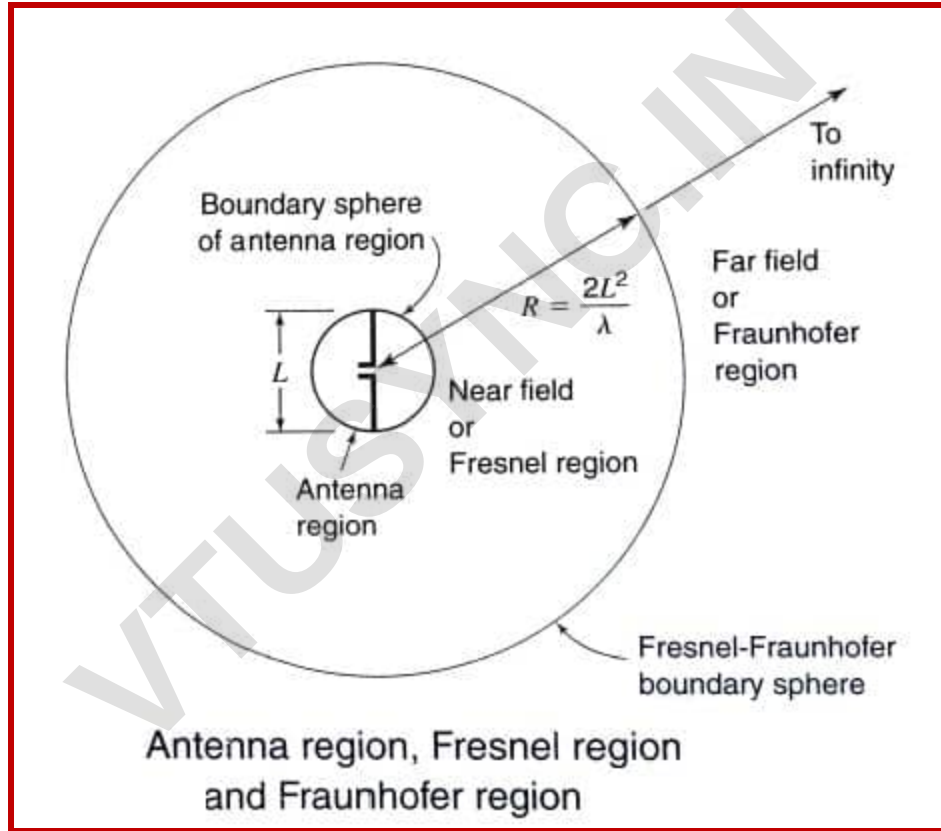
Time: 3 hrs.

Max. Marks: 80

**Note: Answer any FIVE full questions, choosing
ONE full question from each module.**

- 6
- a. Explain co-planar strip line and shielded strip line. (06 Marks)
 - b. Write a note on antenna field zones. (06 Marks)
 - c. An antenna has a field pattern given by $E(\theta) = \cos^2\theta$ for $0 \leq \theta \leq \pi/2$. Find the beam area and directivity. (04 Marks)

Antenna Field Zones



Antenna Field Zones

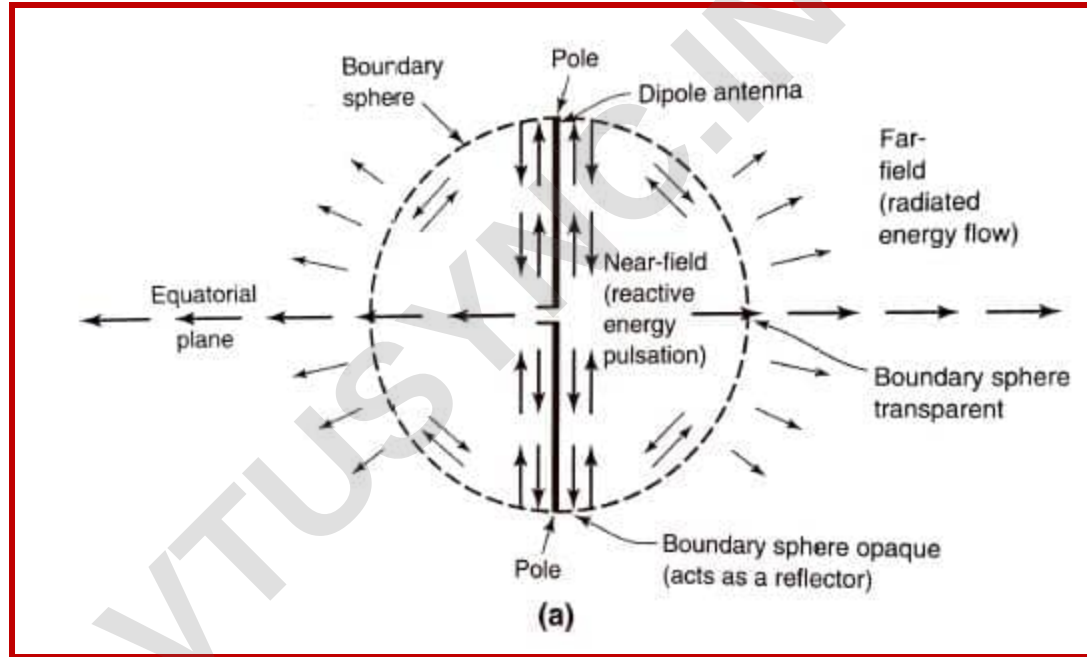
The fields around an antenna may be divided into two principal regions, one near the antenna called the *near field* or *Fresnel zone* and one at a large distance called the *far field* or *Fraunhofer zone*. Referring to Fig. 2-17, the boundary between the two may be arbitrarily taken to be at a radius

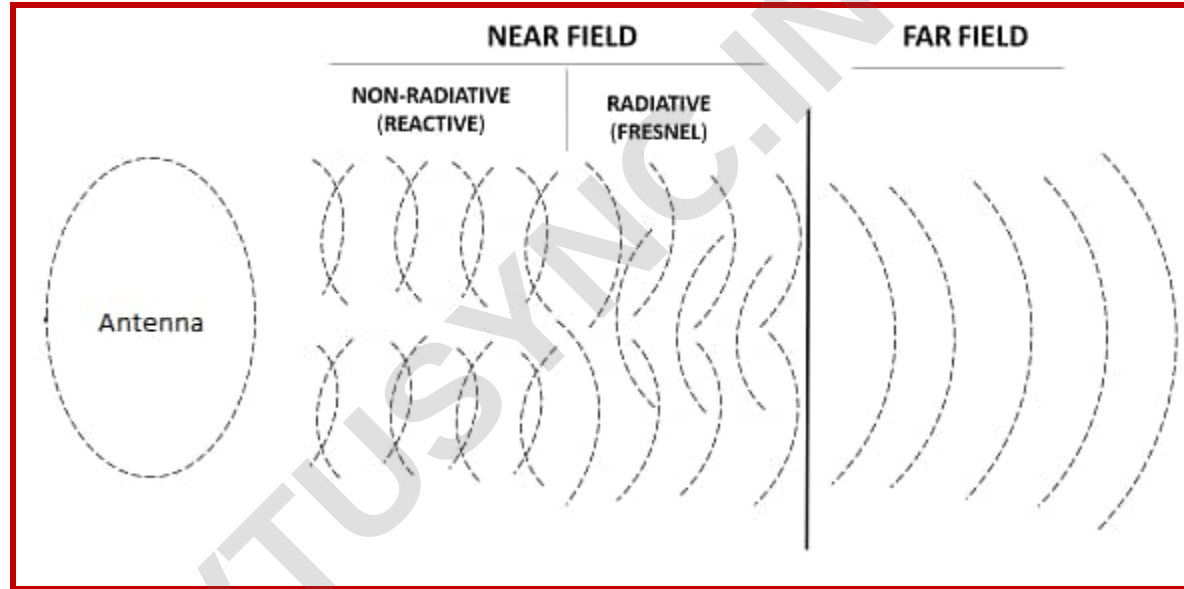
$$R = \frac{2L^2}{\lambda} \quad (\text{m})$$

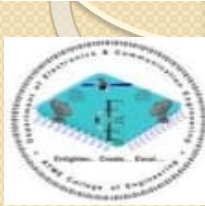
where

L = maximum dimension of the antenna, m

λ = wavelength, m







Near Field (Fresnel Region)

Distance:

The near field region is typically within a few wavelengths of the antenna. The boundary is often considered to be at a distance

$$R < 2L^2 / \lambda$$

where L is the largest dimension of the antenna, and λ is the wavelength.

Field Characteristics:

- The electromagnetic fields in the near field are *complex and reactive*, meaning they have both radiative and non-radiative (evanescent) components.
- The electric and magnetic fields are not orthogonal and vary significantly with distance and position.
- The field strength decreases rapidly with distance, typically as $1/r^3$ or $1/r^2$

Applications:

- Near-field communication (NFC), inductive charging, and certain types of sensing and scanning.
- Used in applications where strong, localized electromagnetic fields are required.

Far Field (Fraunhofer Region)

Distance:

- The far field region is typically at distances greater than

$$R > 2L^2/\lambda$$

- In this region, the distance is large enough that the spherical wavefronts can be approximated as plane waves.

Field Characteristics:

- The electromagnetic fields in the far field are predominantly radiative, meaning they consist mostly of propagating waves.
- The electric and magnetic fields are *orthogonal to each other* and to the direction of propagation.
- The field strength decreases as $1/r$

Applications:

- Used for most antenna applications, including radio, television broadcasting, and radar.
- Far-field measurements are critical for characterizing antenna radiation patterns, gain, and directivity.

Aspect	Near Field	Far Field
Distance	$r < \frac{2D^2}{\lambda}$	$r > \frac{2D^2}{\lambda}$
Field Characteristics	Complex, reactive, non-radiative & radiative	Simple, predominantly radiative, plane waves
Field Strength Decay	Rapid ($\frac{1}{r^3}$ or $\frac{1}{r^2}$)	Slower ($\frac{1}{r}$)
Electric & Magnetic Fields	Not orthogonal, vary with distance	Orthogonal, consistent with plane waves
Applications	NFC, inductive charging, near-field scanning	Broadcasting, communication, far-field antenna measurements

In the far or Fraunhofer region, the measurable field components are transverse to the radial direction from the antenna and all power flow is directed radially outward. In the far field the shape of the field pattern is independent of the distance. In the near or Fresnel region, the longitudinal component of the electric field may be significant and power flow is not entirely radial. In the near field, the shape of the field pattern depends, in general, on the distance.

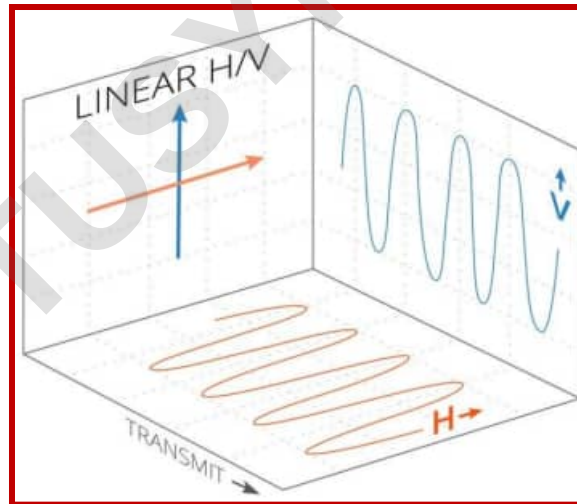
Enclosing the antenna in an imaginary boundary sphere as in Fig. 2-18a it is as though the region near the poles of the sphere acts as a reflector. On the other hand, the waves expanding perpendicular to the dipole in the equatorial region of the sphere result in power leakage through the sphere as if partially transparent in this region.

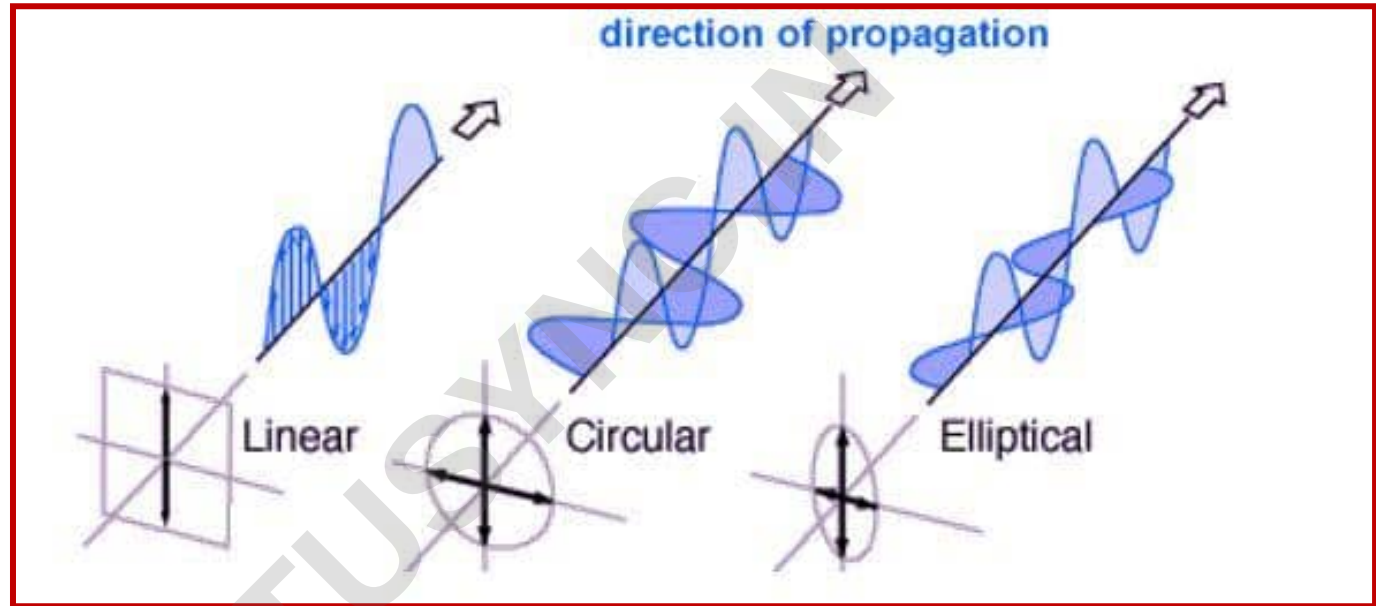
This results in reciprocating (oscillating) energy flow near the antenna accompanied by outward flow in the equatorial region. The outflow accounts for the power radiated from the antenna, while the reciprocating energy represents reactive power that is trapped near the antenna like in a resonator.

Antenna Polarization

The simplest way to describe polarization is the direction in which the electric field of a radio wave oscillates while it propagates through a medium.

This can be visualized by imagining standing directly behind a radio antenna and looking in the direction it is aimed





<https://www.youtube.com/watch?v=Q0qrU4nprB0>

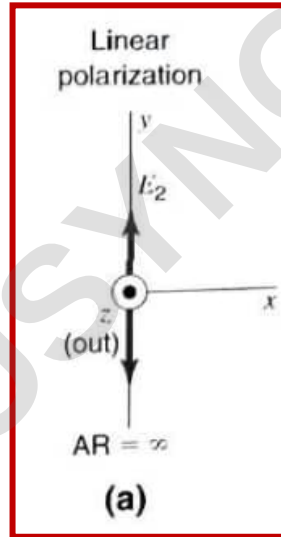
<https://www.youtube.com/watch?v=BycPkRlutqg>

I. Linear Polarization:

- **Linear polarisation:** Linear polarisation is the most common form of antenna polarisation. It is characterised by the fact that all of the radiation is in one plane - hence the term linear.
 - Horizontal polarisation: This form of antenna polarisation has horizontal elements. It picks up and radiates horizontally polarised signals, i.e. electromagnetic waves with the electric field in the horizontal plane.
 - Vertical polarisation: This form of antenna is typified by the vertical elements within the antenna. It could be a single vertical element.
 - Slant polarisation: This is a form of radio antenna polarisation that is at an angle to the horizontal or vertical planes. In this way both vertical and horizontally polarised antennas are able to receive the signal.

$$E_y = E_2 \sin(\omega t - \beta z)$$

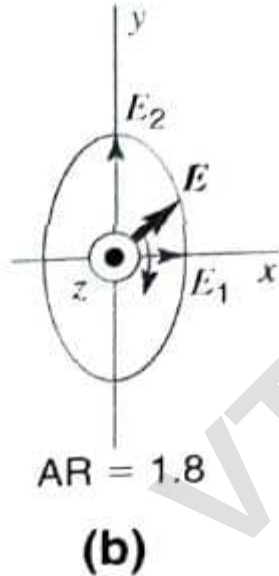
(1)



If $E_1 = 0$, the wave is linearly polarized in the y direction. If $E_2 = 0$, the wave is linearly polarized in the x direction.

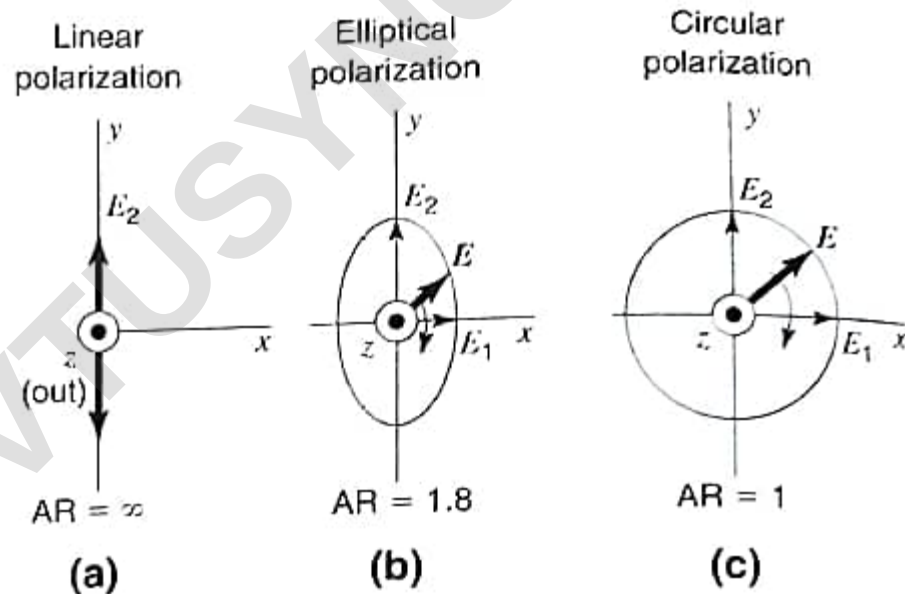
2. Elliptical Polarization:

Elliptical
polarization



In general, the electric field of a wave traveling in the z direction may have both a y component and an x component, as suggested in Fig. 2-21b. In this more general situation, with a phase difference δ between the components, the wave is said to be *elliptically polarized*. At a fixed value of z the electric vector $\mathbf{E}$ rotates as a function of time, the tip of the vector describing an ellipse called the *polarization ellipse*. The ratio of the major to minor axes of the polarization ellipse is called the *Axial Ratio* (AR). Thus, for the wave Fig. 2-21b, $AR = E_2/E_1$.

Two extreme cases of elliptical polarization correspond to *circular polarization*, as in Fig. 2-21c, and *linear polarization*, as in Fig. 2-21a. For circular polarization $E_1 = E_2$ and $AR = 1$, while for linear polarization $E_1 = 0$ and $AR = \infty$.



δ = time-phase angle by which E_y leads E_x



Combining (2) and (3) gives the instantaneous total vector field $\mathbf{E}$:

$$\mathbf{E} = \hat{x}E_1 \sin(\omega t - \beta z) + \hat{y}E_2 \sin(\omega t - \beta z + \delta) \quad (4)$$

At $z = 0$, $E_x = E_1 \sin \omega t$ and $E_y = E_2 \sin(\omega t + \delta)$. Expanding E_y yields

$$E_y = E_2(\sin \omega t \cos \delta + \cos \omega t \sin \delta) \quad (5)$$

From the relation for E_x we have $\sin \omega t = E_x/E_1$ and $\cos \omega t = \sqrt{1 - (E_x/E_1)^2}$. Introducing these in (5) eliminates ωt , and, on rearranging, we obtain

$$\frac{E_x^2}{E_1^2} - \frac{2E_x E_y \cos \delta}{E_1 E_2} + \frac{E_y^2}{E_2^2} = \sin^2 \delta \quad (6)$$

or

$$aE_x^2 - bE_x E_y + cE_y^2 = 1 \quad (7)$$

where

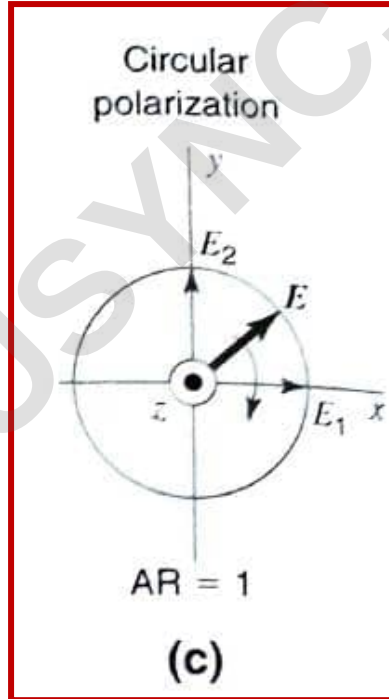
$$a = \frac{1}{E_1^2 \sin^2 \delta} \quad b = \frac{2 \cos \delta}{E_1 E_2 \sin^2 \delta} \quad c = \frac{1}{E_2^2 \sin^2 \delta}$$

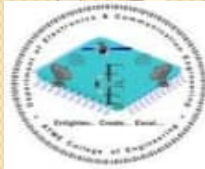
Equation (7) describes a (polarization) ellipse, as in Fig. 2-22. The line segment OA is the semimajor axis, and the line segment OB is the semiminor axis. The tilt angle of the ellipse is τ . The axial ratio is

$$AR = \frac{OA}{OB} \quad (1 \leq AR \leq \infty) \quad \text{Axial Ratio} \quad (8)$$

3. Circular Polarization:

If $E_1 = E_2$ and $\delta = \pm 90^\circ$, the wave is circularly polarized. When $\delta = +90^\circ$, the wave is *left circularly polarized*, and when $\delta = -90^\circ$, the wave is *right circularly polarized*.





Thank you